Chapter 2

Network Models

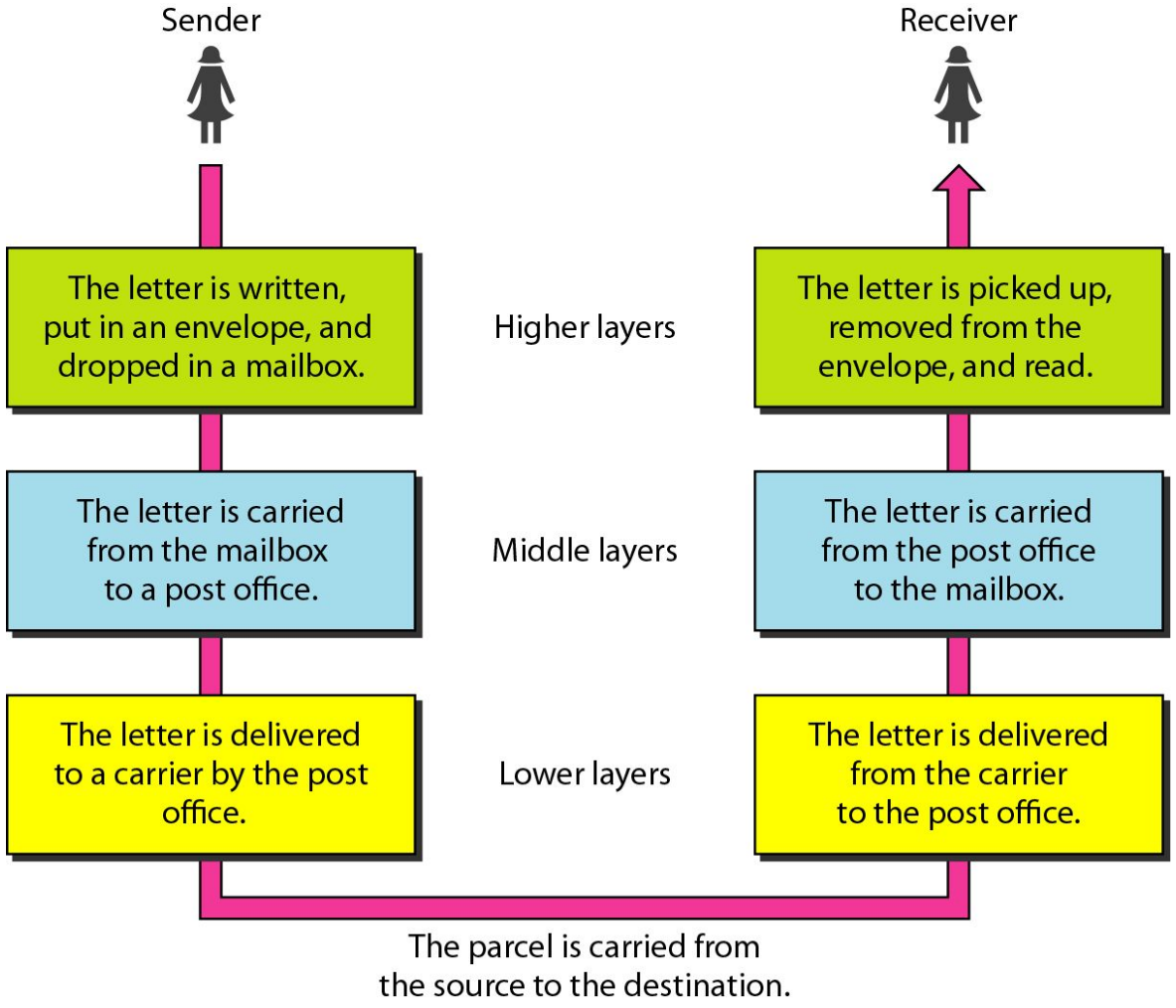
2-1 LAYERED TASKS

We use the concept of layers in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.

Topics discussed in this section:

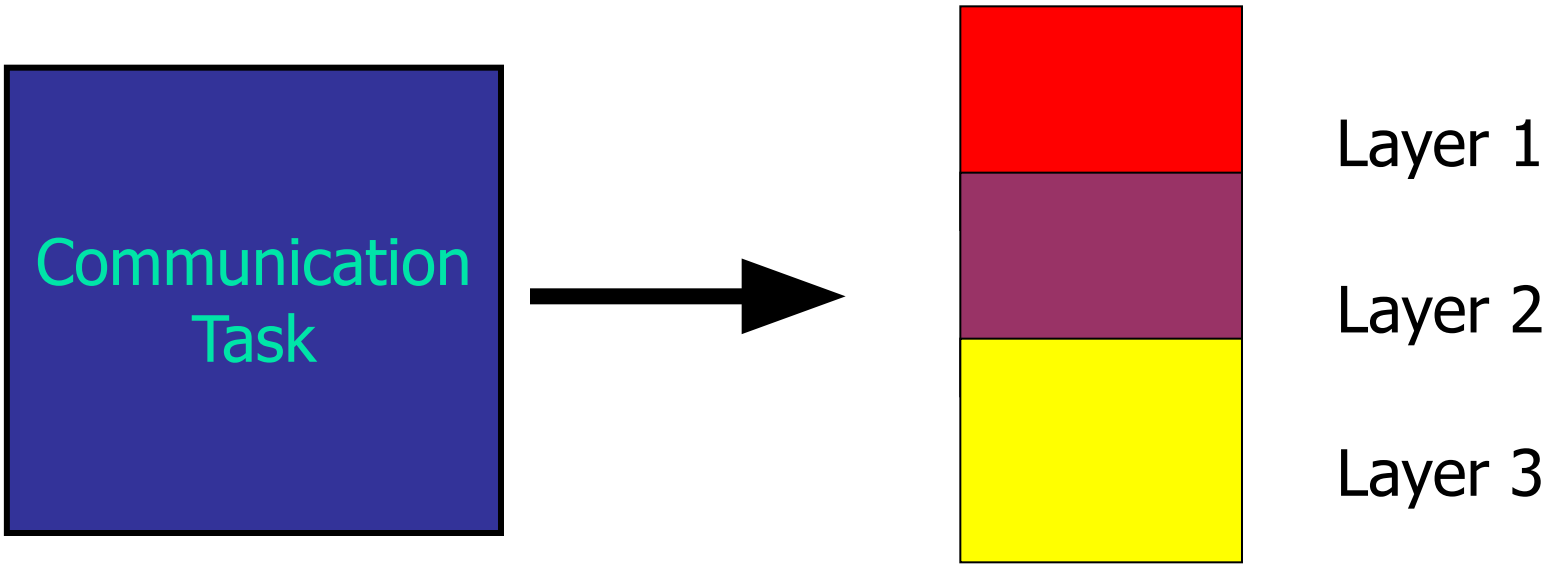
Sender, Receiver, and Carrier
Hierarchy

Figure 2.1 Tasks involved in sending a letter



Layering

- Task of communication broken up into layers



Layers involving calling a friend

Where is the Café?

Content layer

Conversation Protocol Suite
1. Use a Common Language
2. Wait Your Turn
3. Signal When Finished

Rules layer

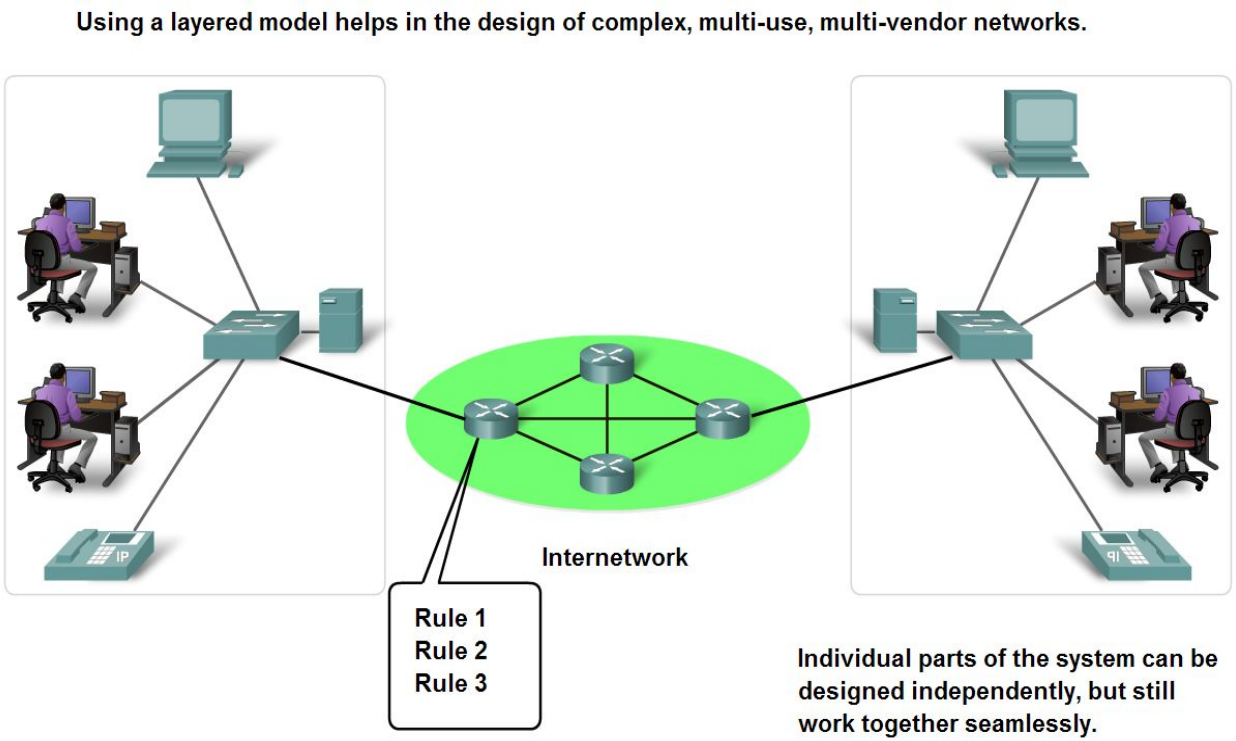


Physical layer



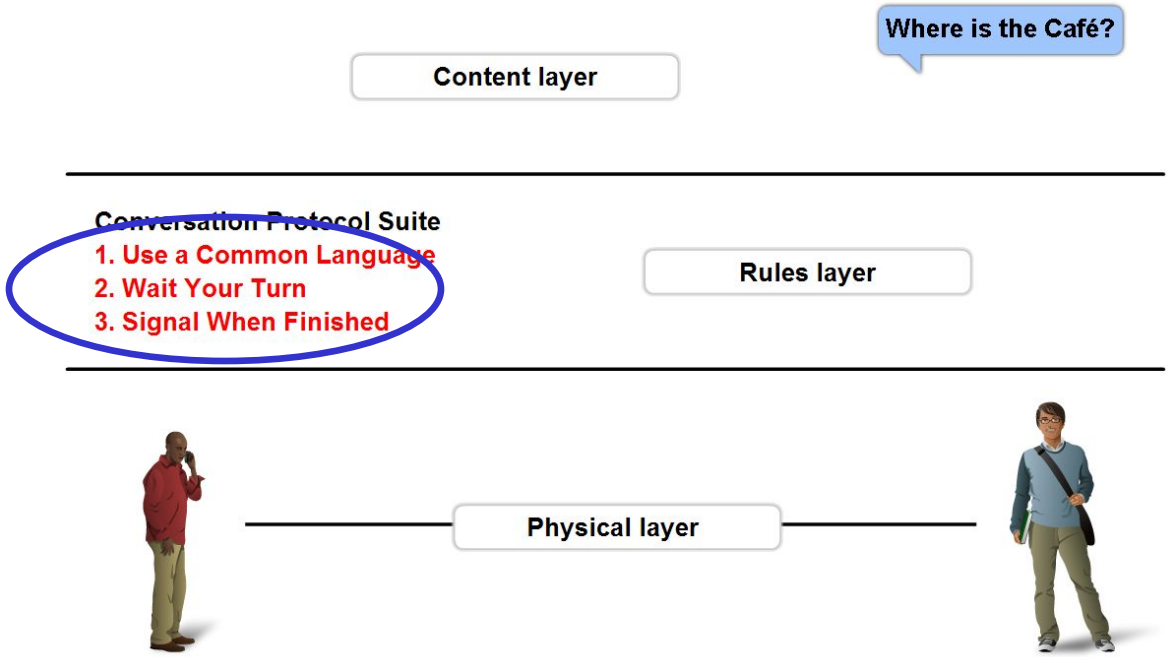
Benefits of using a layered model

- To find out the fault easily.
- Changes in one layer do not affect other layers.
- Have defined information that they act upon.



Rules that govern communications

- A protocol is a set of predetermined rules.
- Defines:
  - What is communicated??
  - How it is communicated??
  - When it is communicated??



Protocols

- Describe processes such as:

The format or structure of the message. What?

The method by which networking devices share information about pathways with other network How?

How and when error and system messages are passed between devices. How/When?

The setup and termination of data transfer sessions. What/When/How?

Protocols and Standards

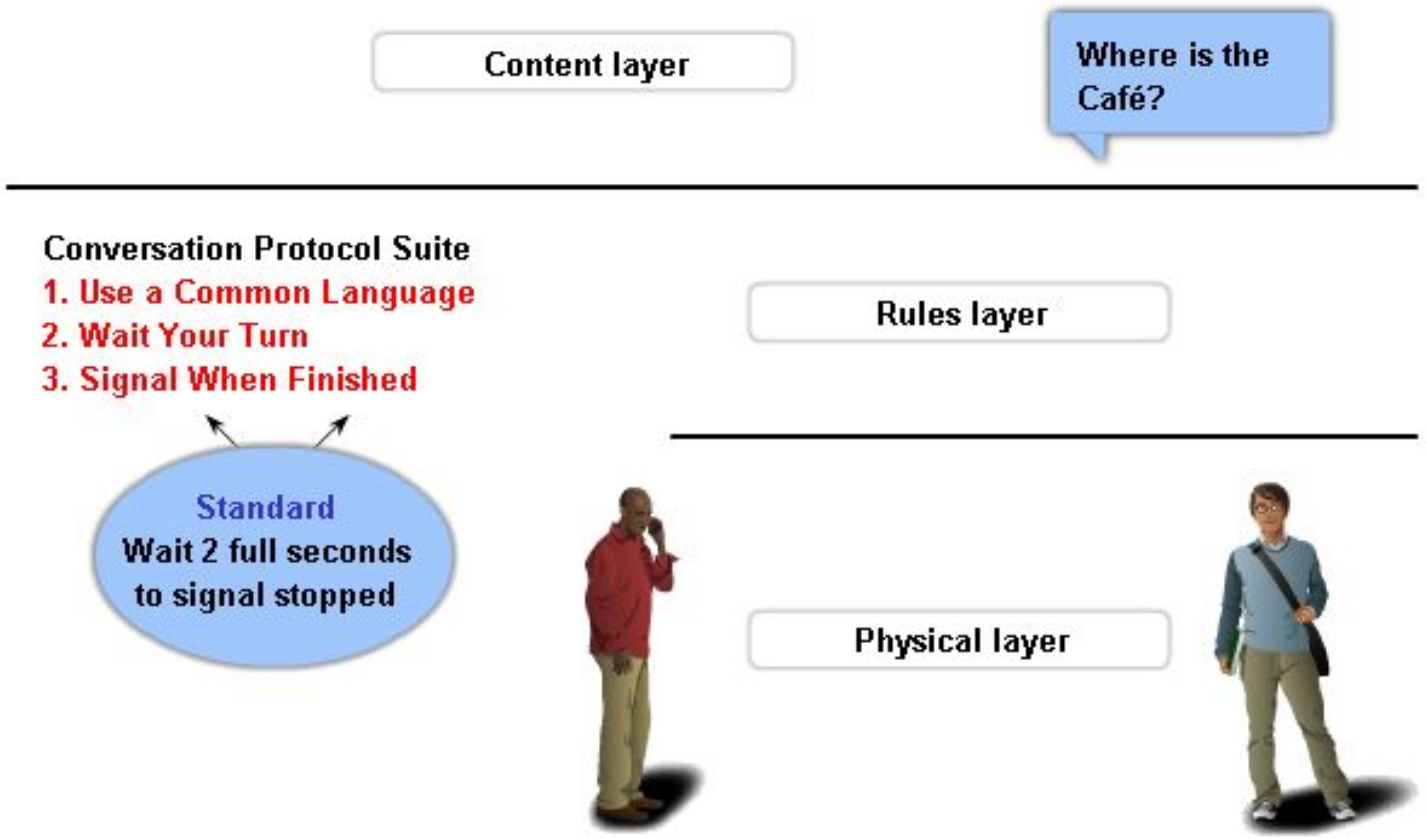
- **Protocols**
- **Standards**
- **Standards Organizations**
- **Internet Standards**

Standards

- Endorsed by the networking industry and approved by a standards organization.
- Benefits:
  - Create and maintain an open and competitive market.
  - Ensured greater compatibility and interoperability.
- Categories
- **De facto:** Standards that have not been approved by an organized body but have been adopted as standards through widespread use
- **De jure:** Those standards that have been legislated by an officially recognized body

Standards and Protocols

Standards are protocols and agreements that are widely used and accepted.



Standard Organizations

- International Organization for Standardization (**ISO**)
- Institute of Electrical and Electronic Engineers (**IEEE**)
- American National Standards Institute (**ANSI**)
- Telecommunications Industry Association (**TIA**)
- The Internet Engineering Task Force (**IETF**)
- International Telecommunications Union – Telecommunication Standards Sector (**ITU-T**)

Communication Process

Layered standards:

- **OSI Reference model**
  - De Jure Standard
- **TCP/IP Protocol Model**
  - Open De Facto Standard
  - Governed by IETF Working Groups

2-2 THE OSI MODEL

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

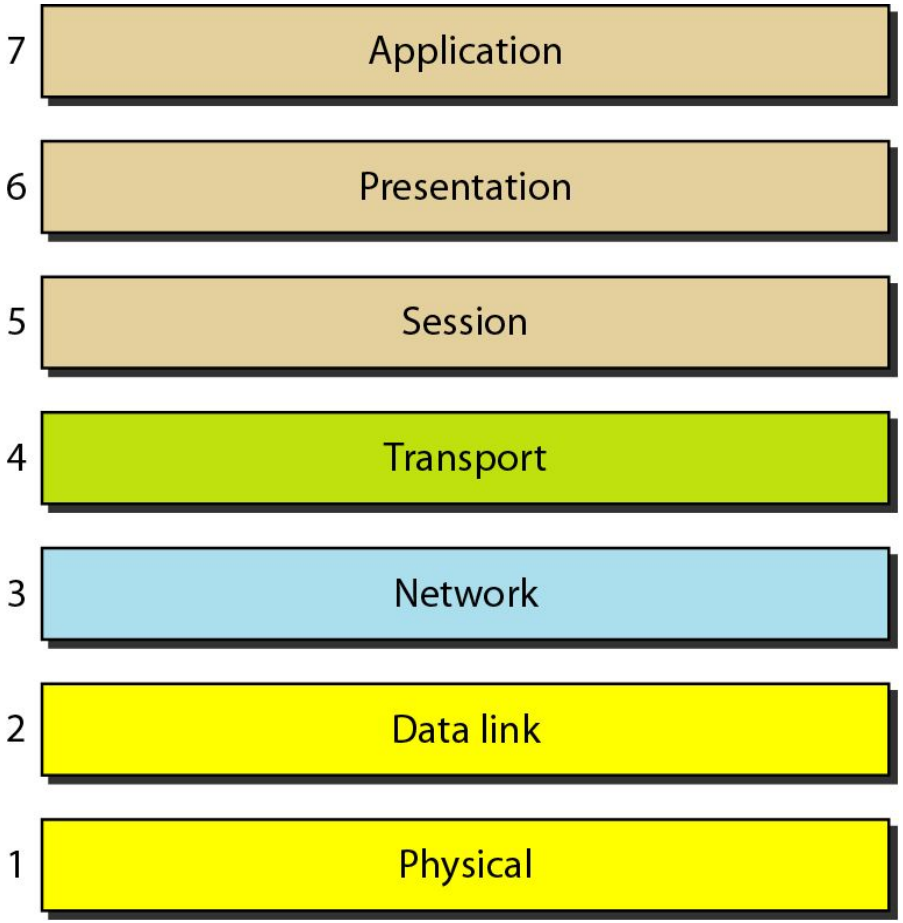
Topics discussed in this section:

Layered Architecture
Peer-to-Peer Processes
Encapsulation

Note

ISO is the organization.
OSI is the model.

Figure 2.2 *Seven layers of the OSI model*



OSI Model- 7 Layers

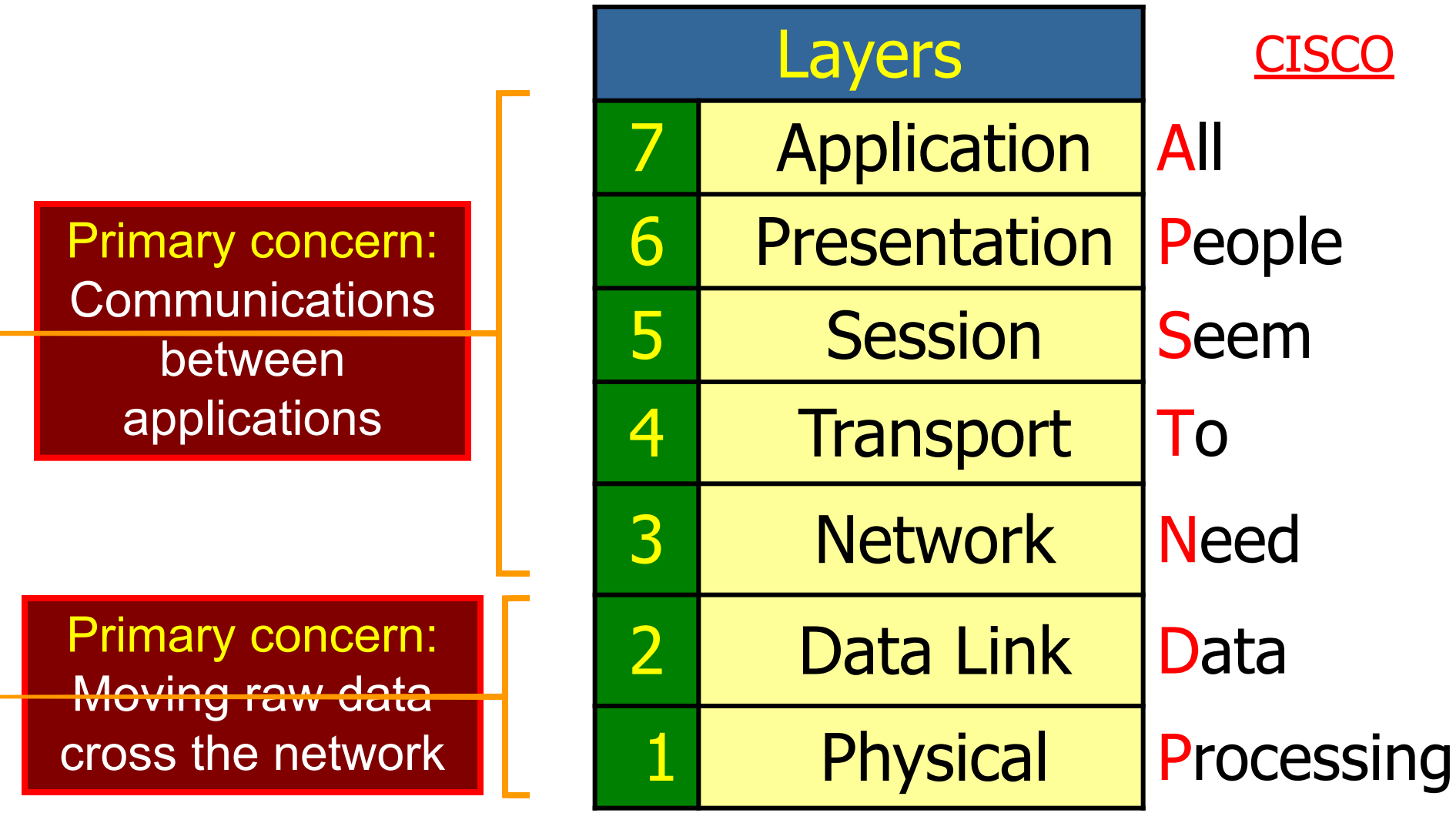
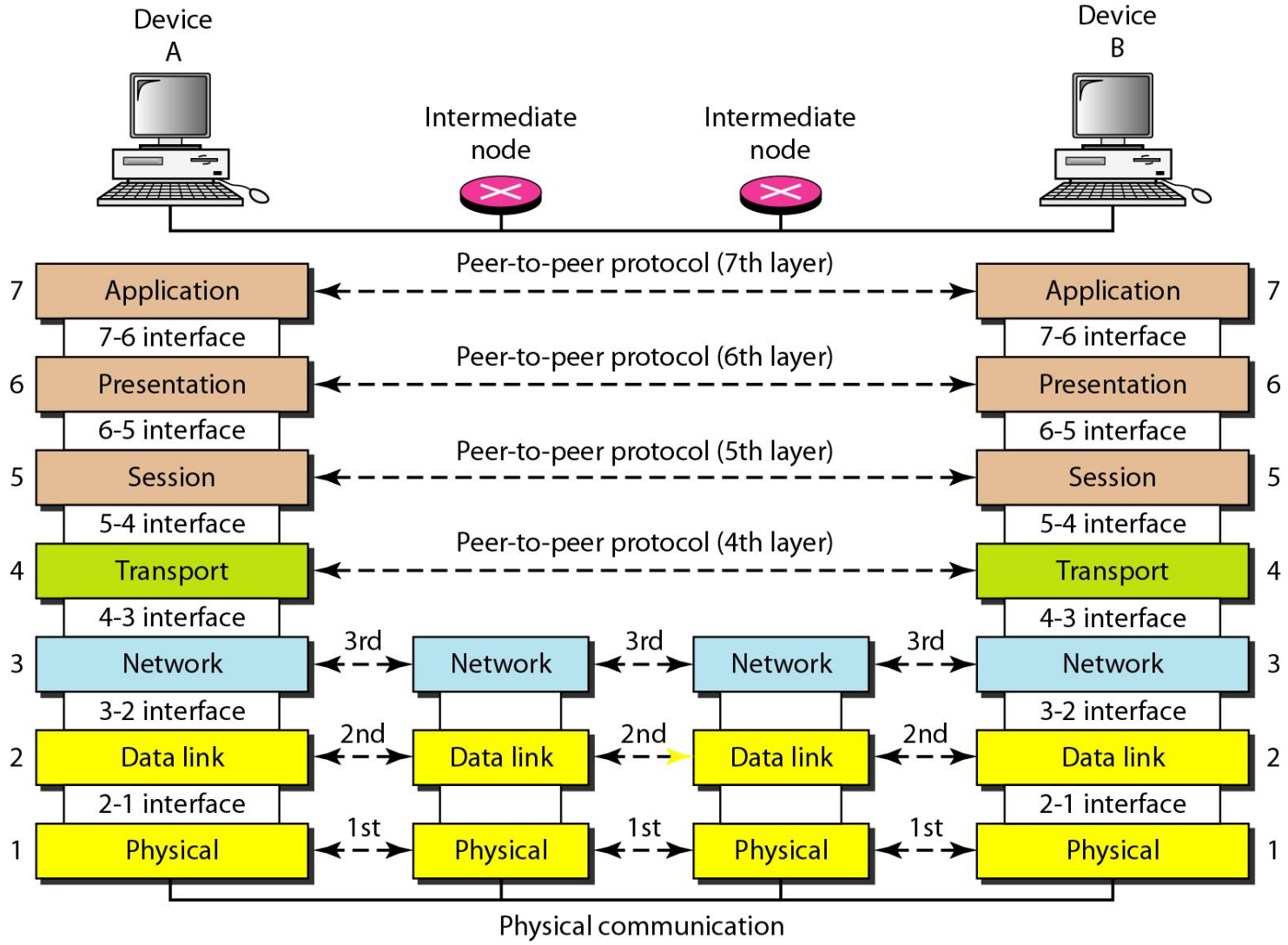
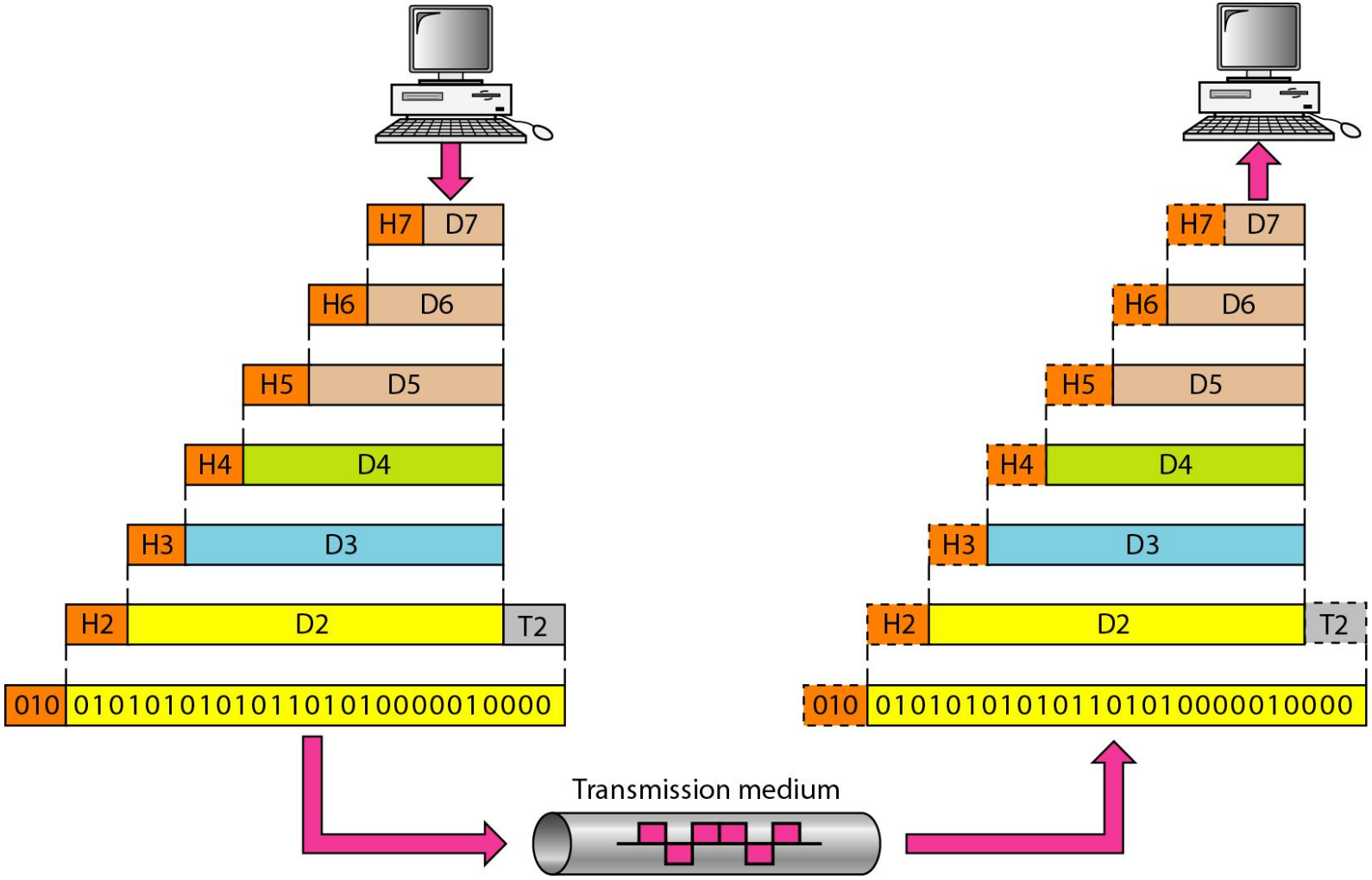


Figure 2.3 *The interaction between layers in the OSI model*



An exchange using the OSI model



2-3 LAYERS IN THE OSI MODEL

In this section we briefly describe the functions of each layer in the OSI model.

Topics discussed in this section:

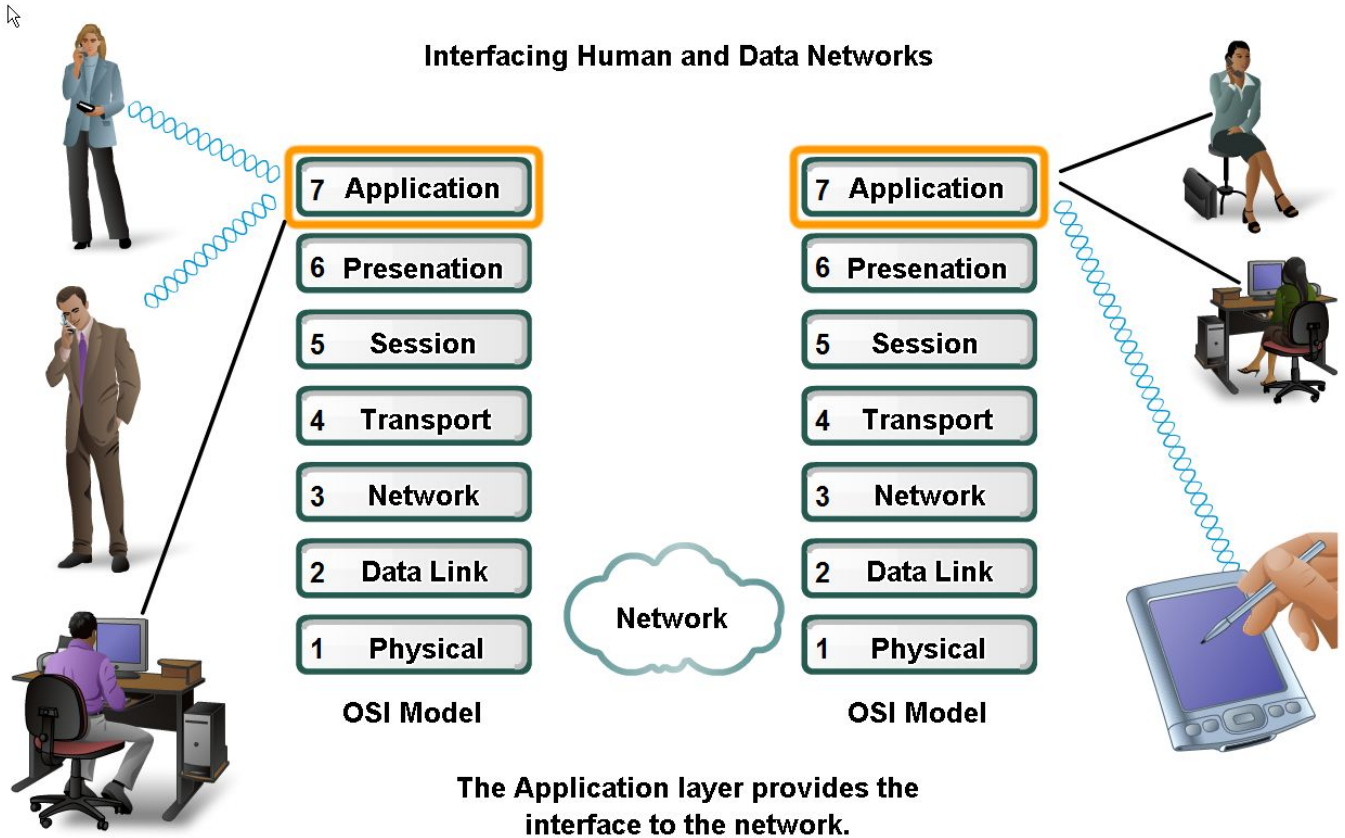
Physical Layer
Data Link Layer
Network Layer
Transport Layer
Session Layer
Presentation Layer
Application Layer

APPLICATION LAYER

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Applications

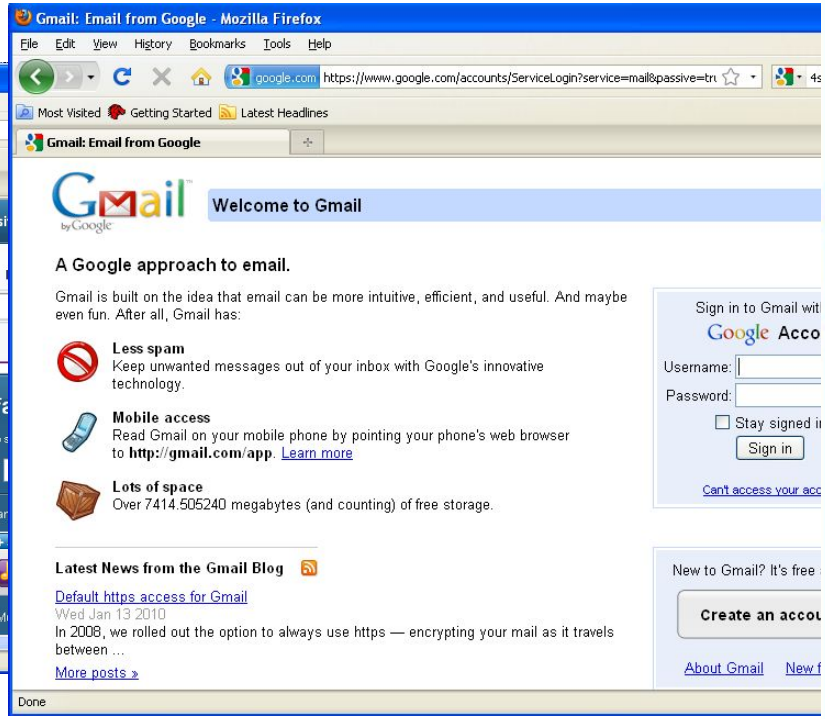
- The Interface Between Human and Data Networks
- Responsible for providing services to the user.



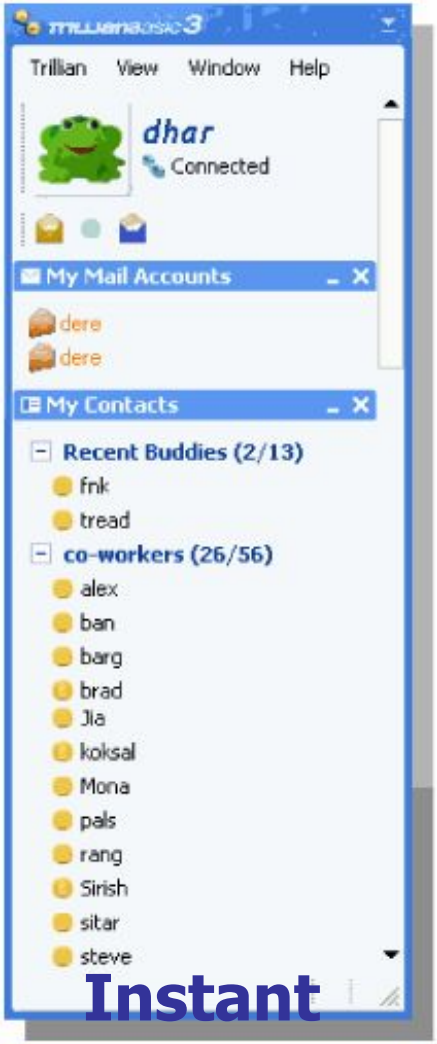
Applications in Application layer



Web Page

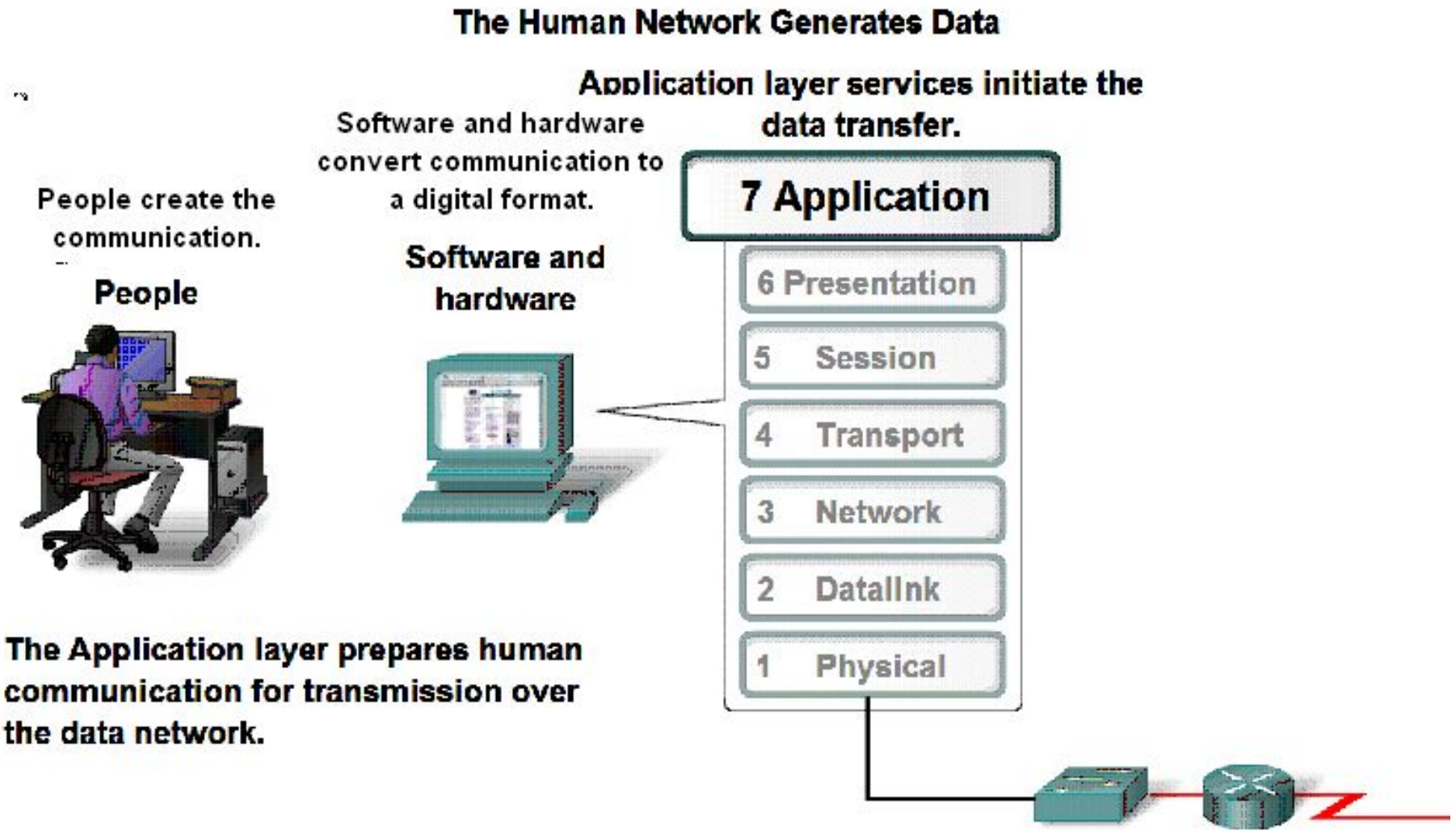


Email

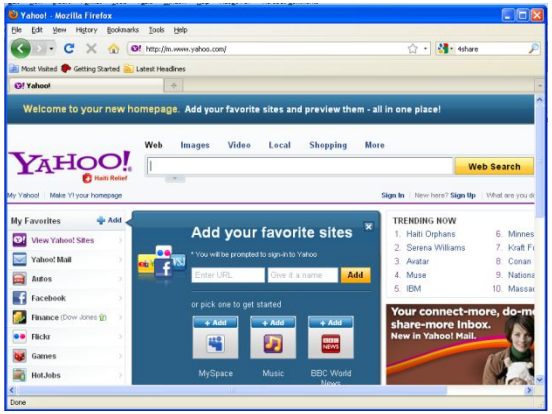


Instant messaging

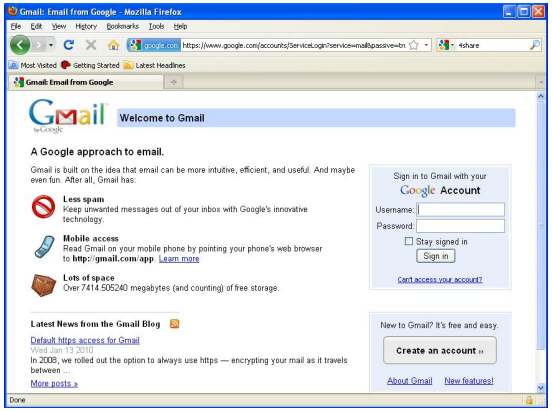
Application Layer



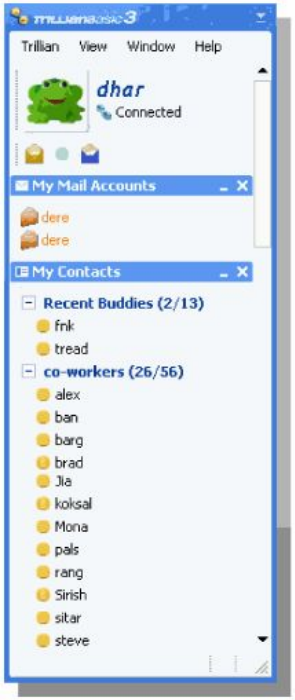
Applications in Application layer



Web Page



Email

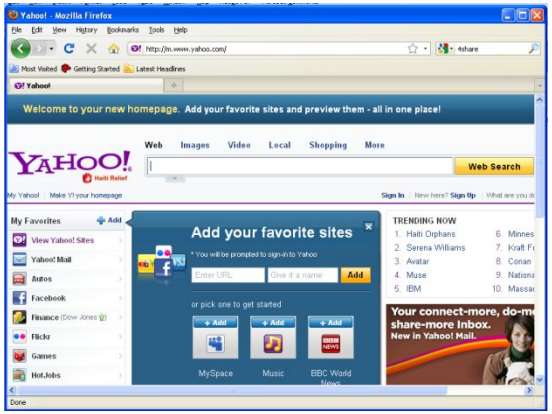


Instant messaging

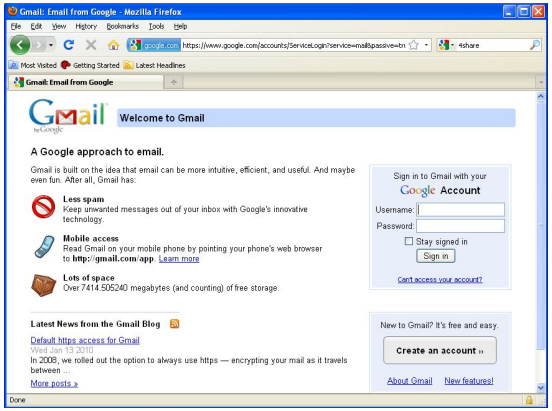


To Presentation Layer

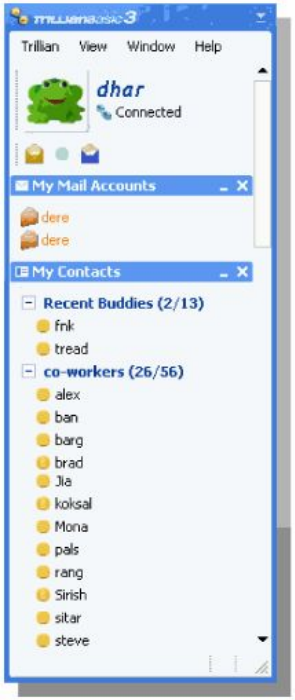
Application layer



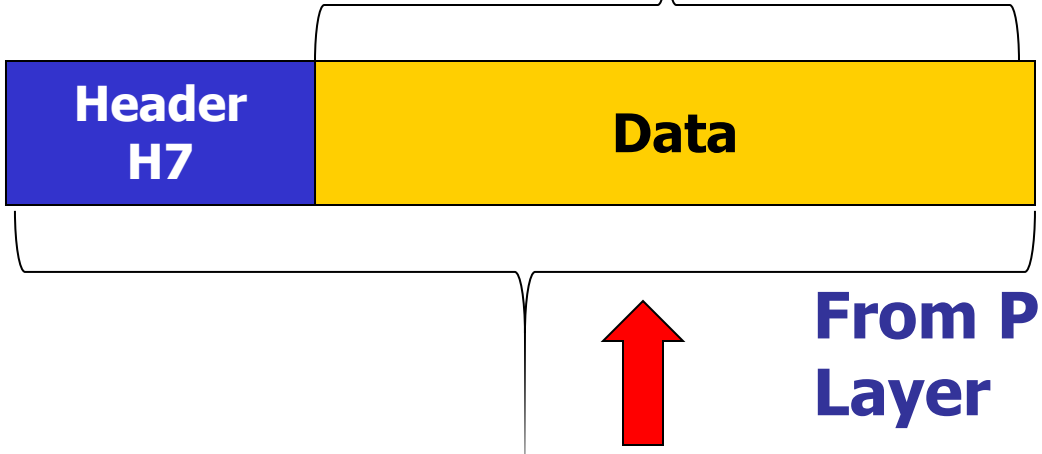
Web Page



Email

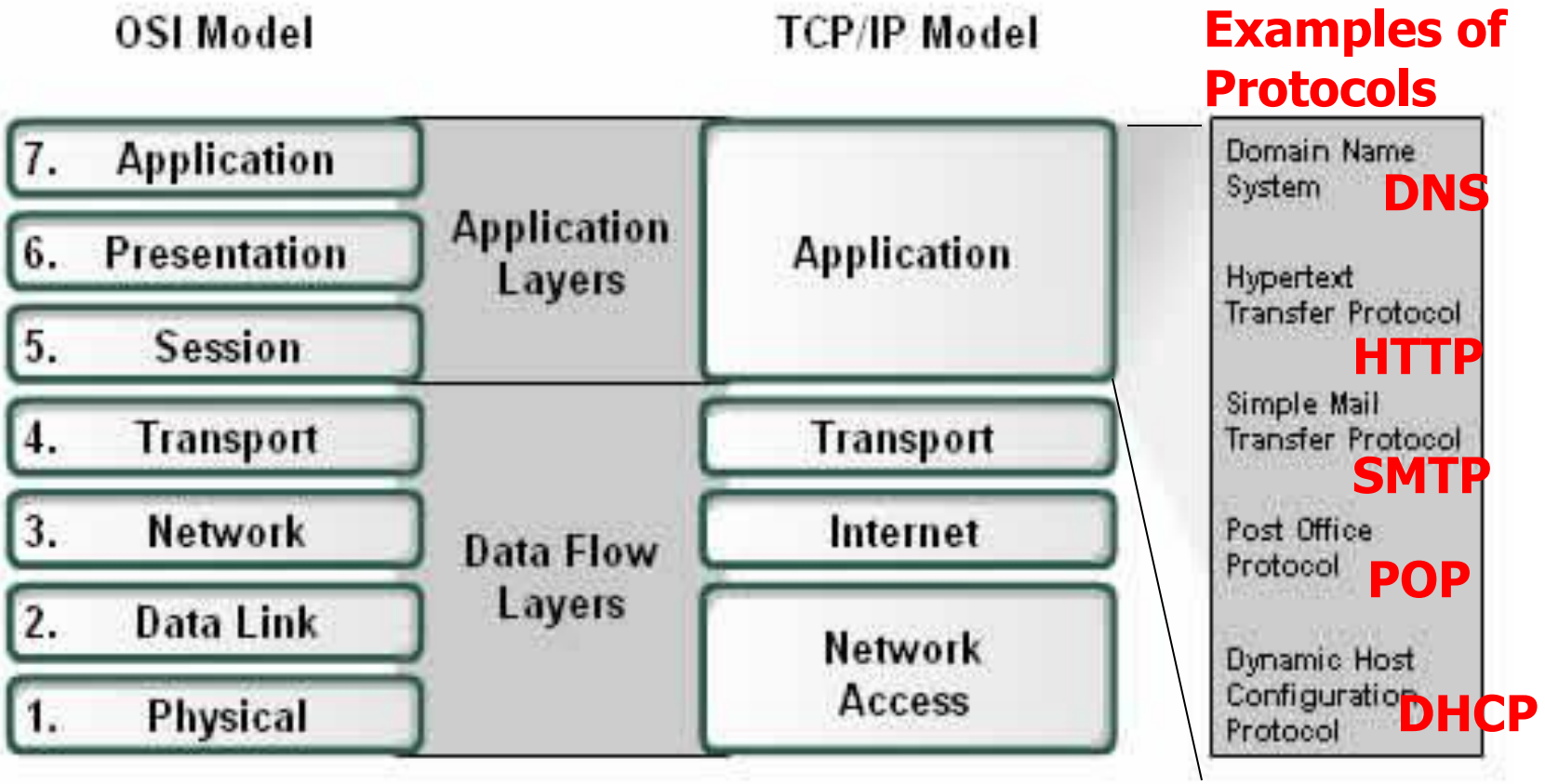


Instant messaging



From Presentation Layer

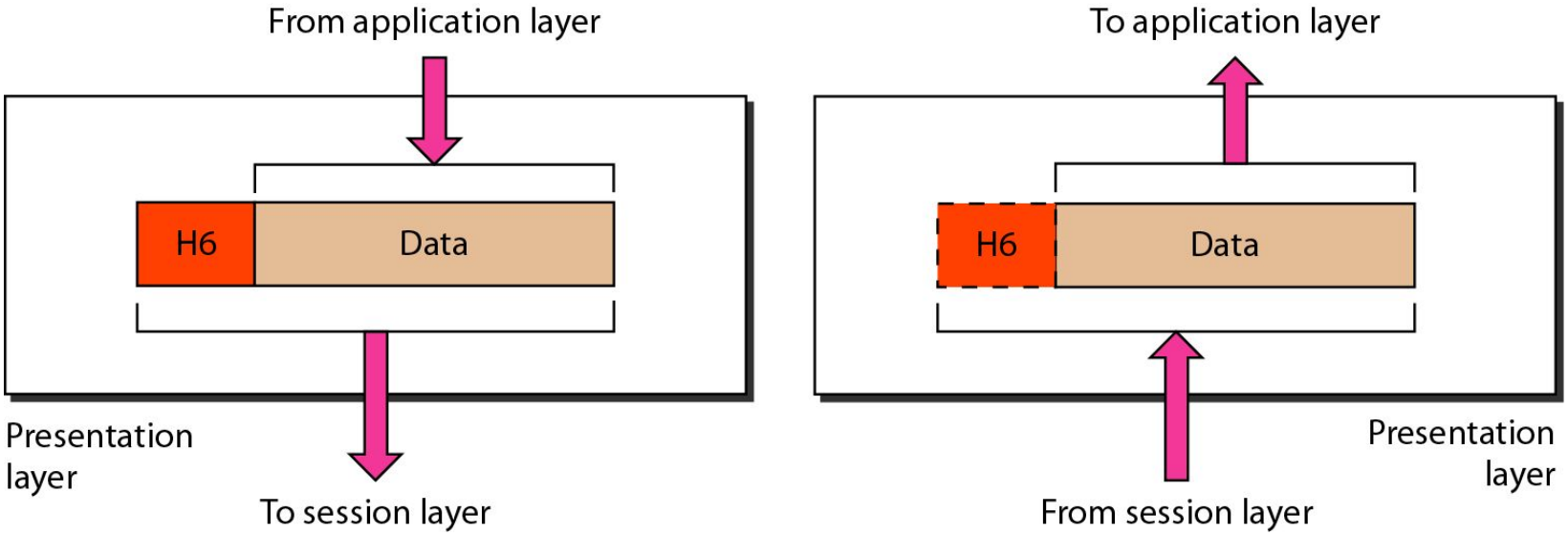
Application Layer



PRESENTATION LAYER

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Presentation layer



The presentation layer is responsible for translation, compression, and encryption.

Presentation Layer

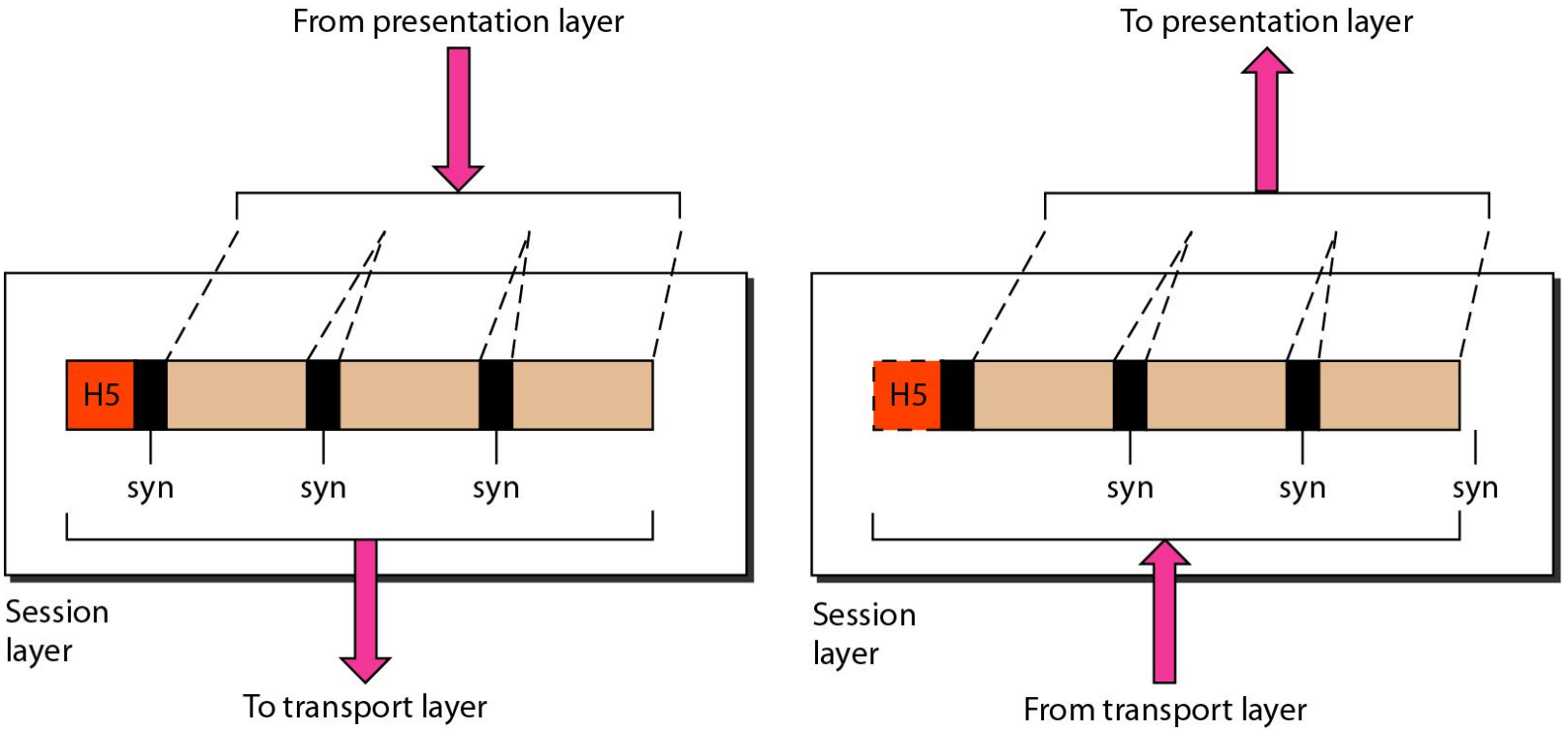
- 3 primary functions:
  - Coding and conversion
  - Compression of the data
  - Encryption of the data

- Presentation layer implementations are not typically associated with a particular protocol stack.

SESSION LAYER

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Session layer



The session layer is responsible for dialog control and synchronization.

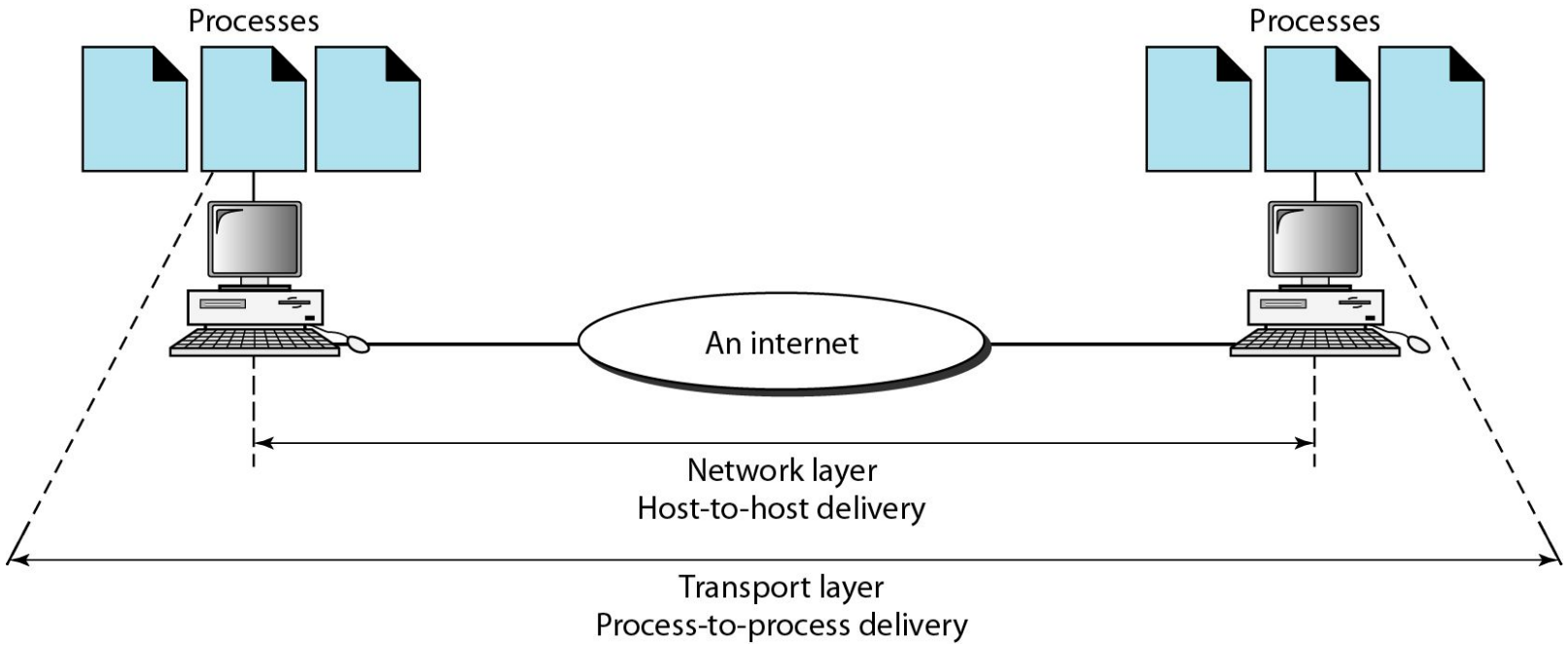
Session Layer

- It handles the exchange of information
  - to initiate dialogs,
  - keep them active, and
  - to restart sessions that are disrupted or idle for a long period of time
- Most applications, like web browsers or e-mail clients, incorporate functionality of the OSI layers 5, 6 and 7.

TRANSPORT LAYER

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Transport layer



The transport layer is responsible for the delivery of a message from one process to another.

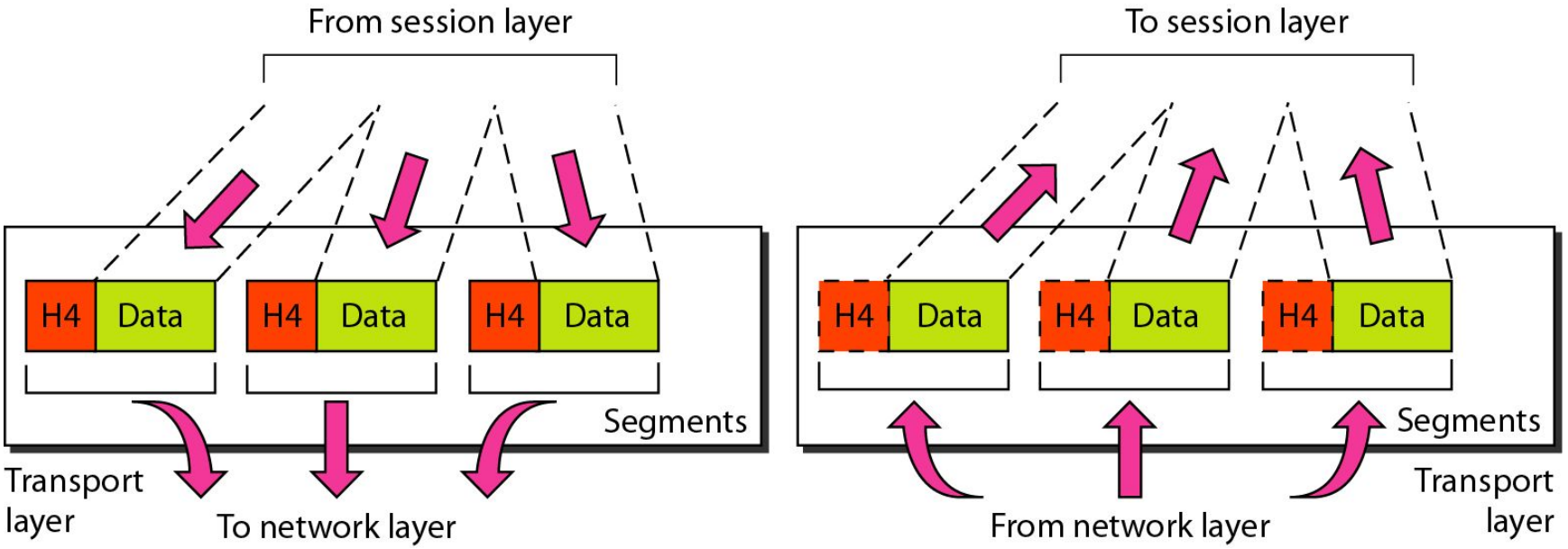
Functions– Transport Layer

- Segmentation and Reassembly
- Adds Port Address and Sequence Number
- Connection Control
- Flow and Error Control
- Multiplexing

Transport Layer PDU is called **Segments**.

Common Protocol used in Transport Layer is **TCP**

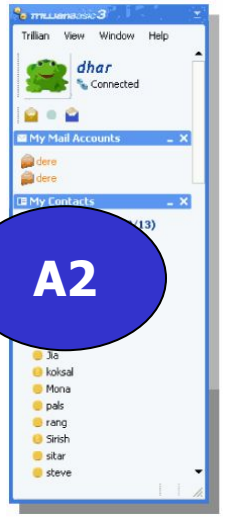
Transport layer



- Segments data received from application layer into small parts.
- Transport Layer Protocol Data Unit is called **Segments**.

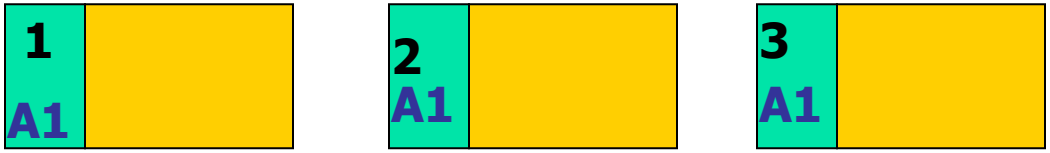
Function: Segmentation





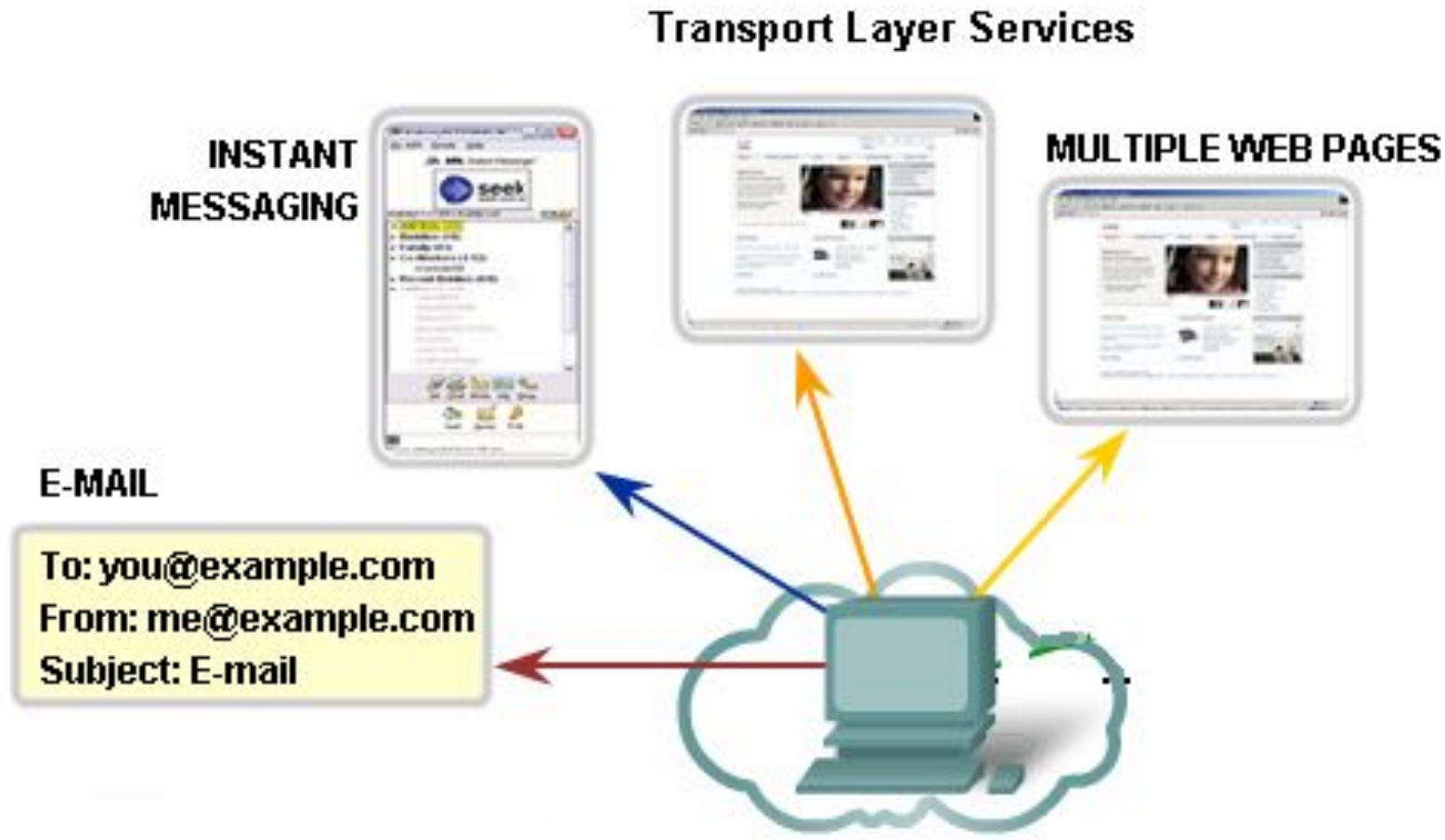
Data from Application layer

Received by Transport Layer



1. **Segments into small parts**
2. **Add a number to identify the application.**
3. **Add a number sequence the segmented parts.**

Identifying Different Applications



- Port Numbers

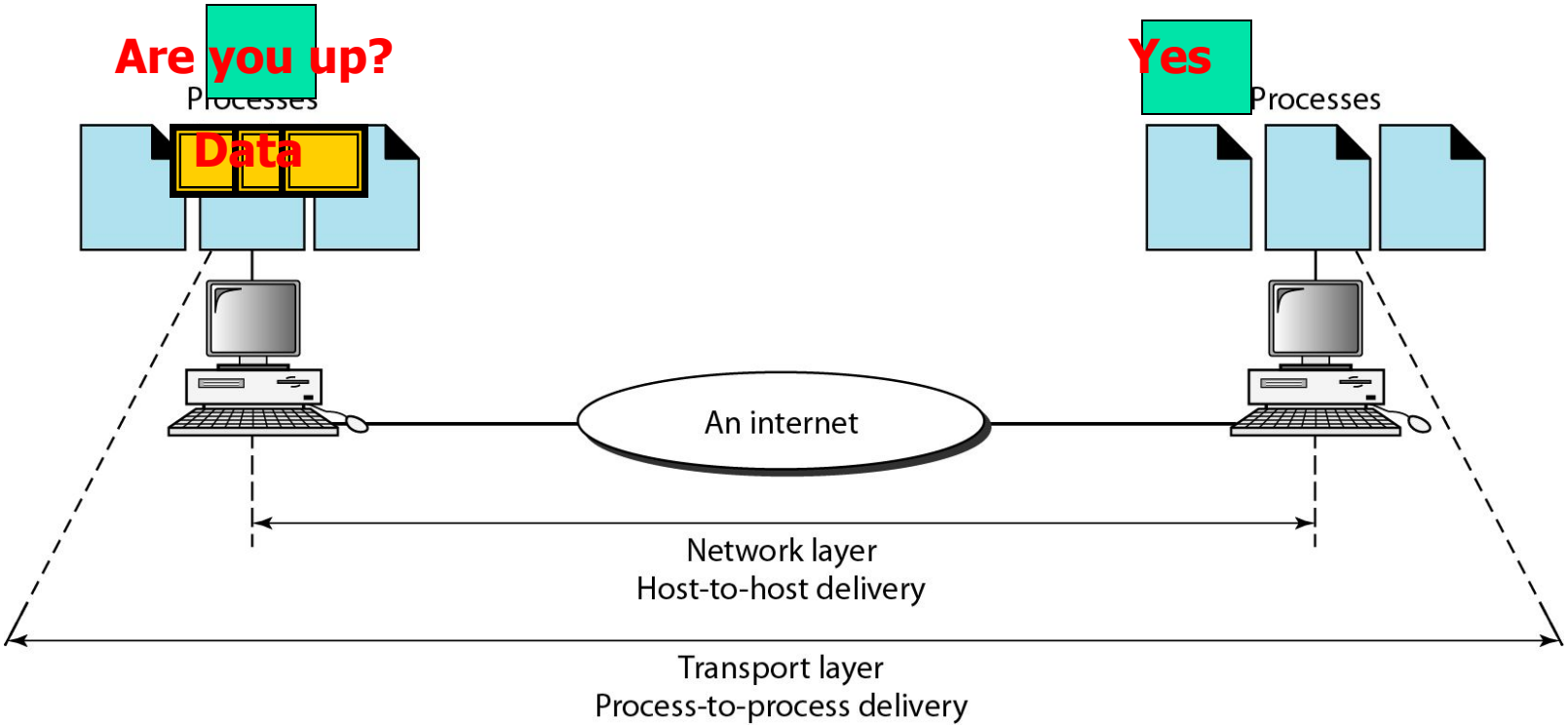
Port Address

- To define multiple processes running in a computer.
- 16-bit in length

80

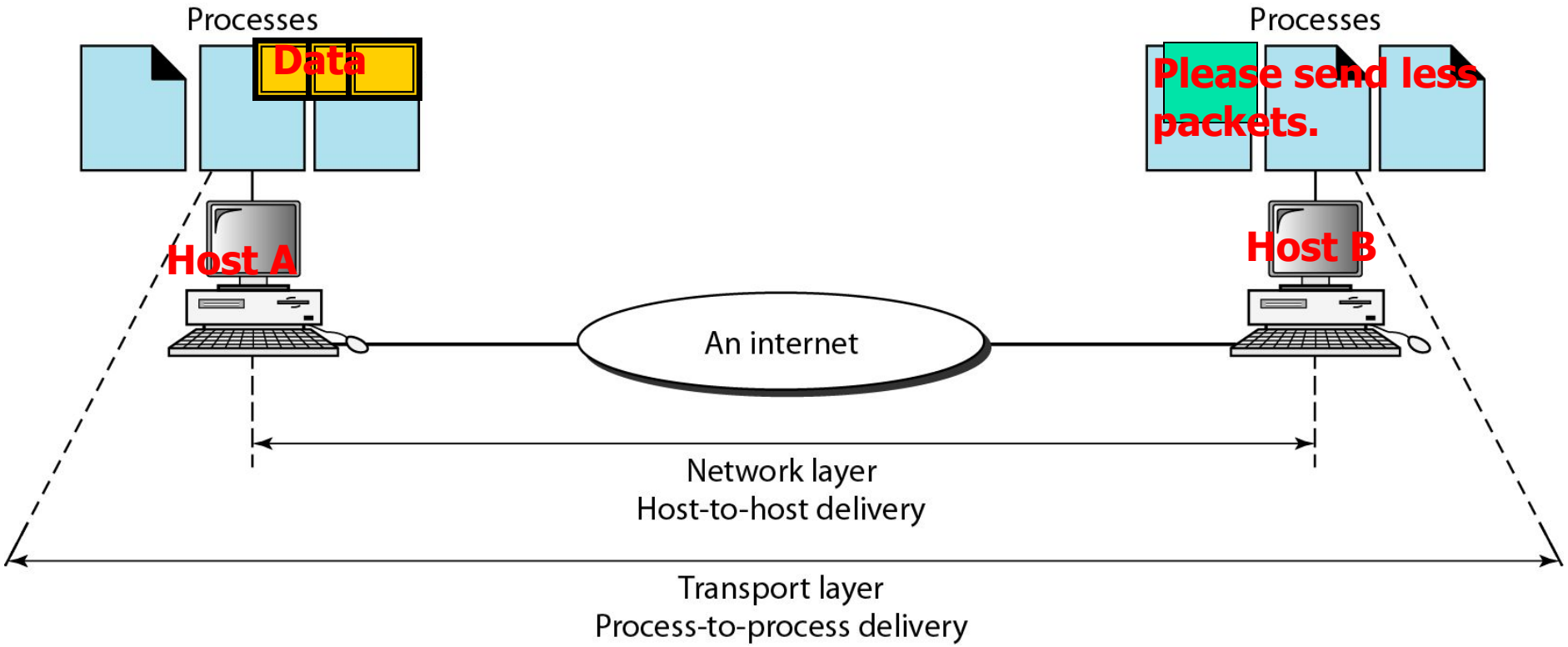
A 16-bit port address represented as one single number.

Function: Connection Control

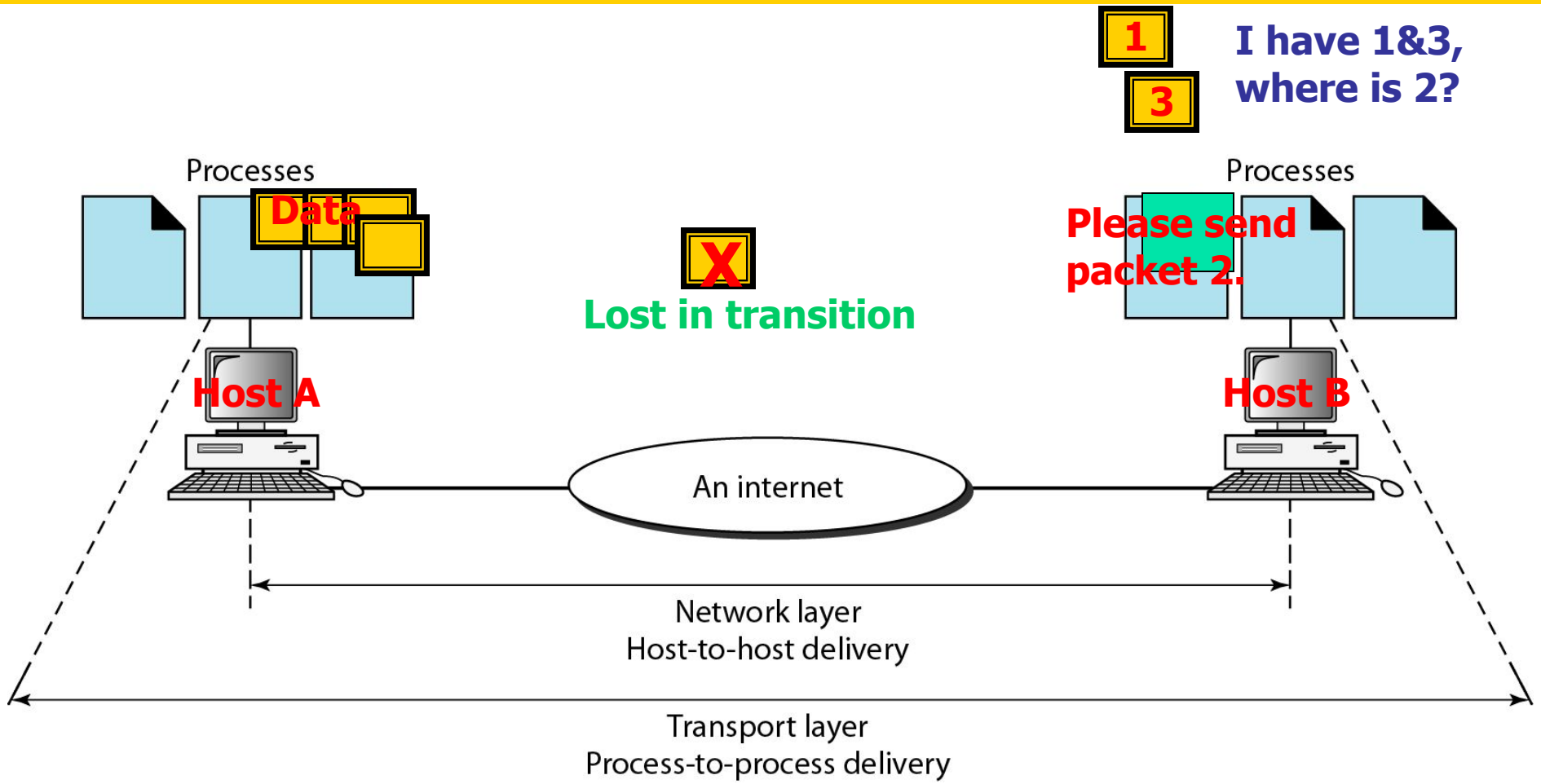


Function: Flow Control

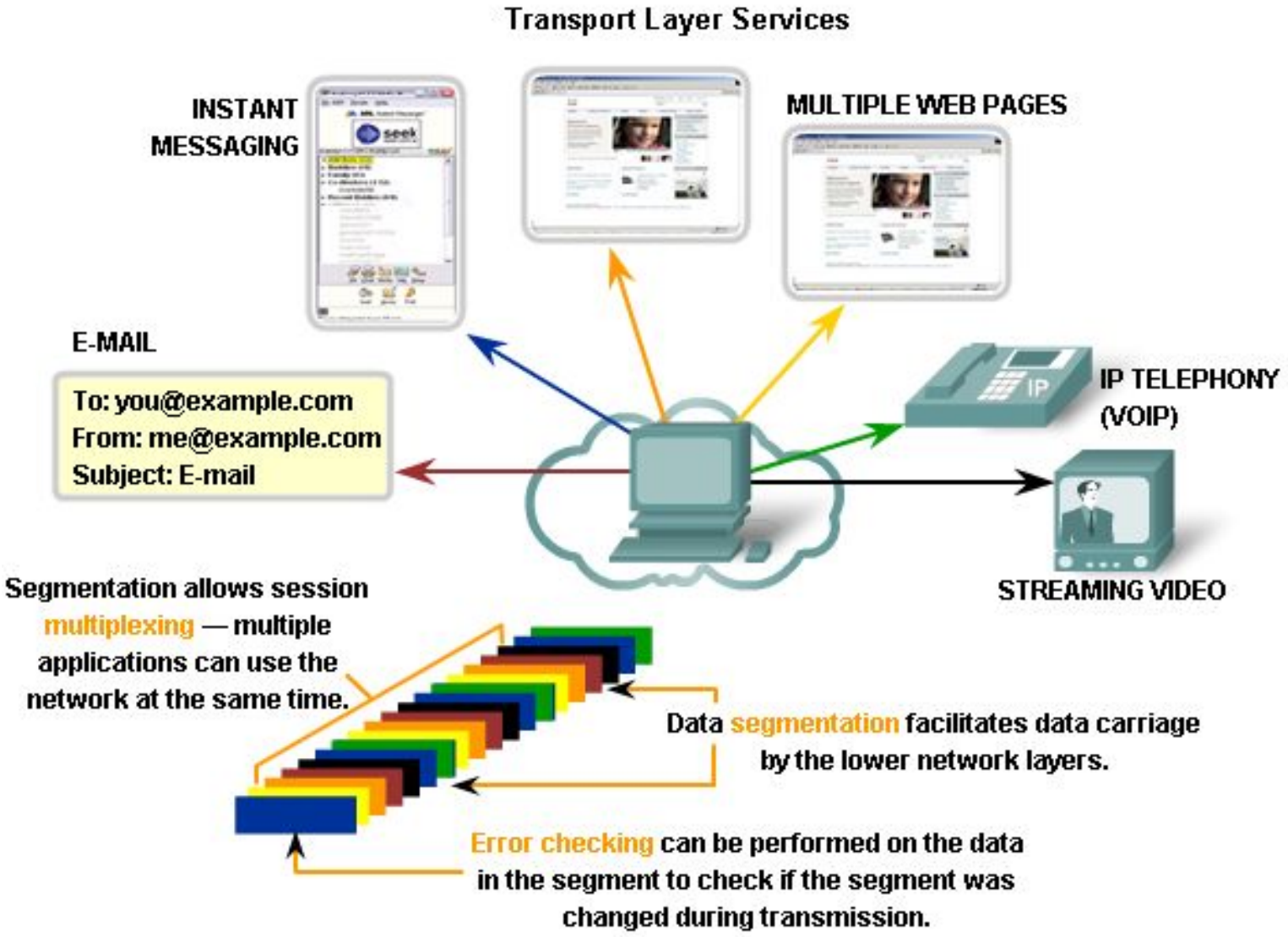
- **Host B has too many packets to process.**
- **Buffer to store incoming packets overflows**



Function: Error Control



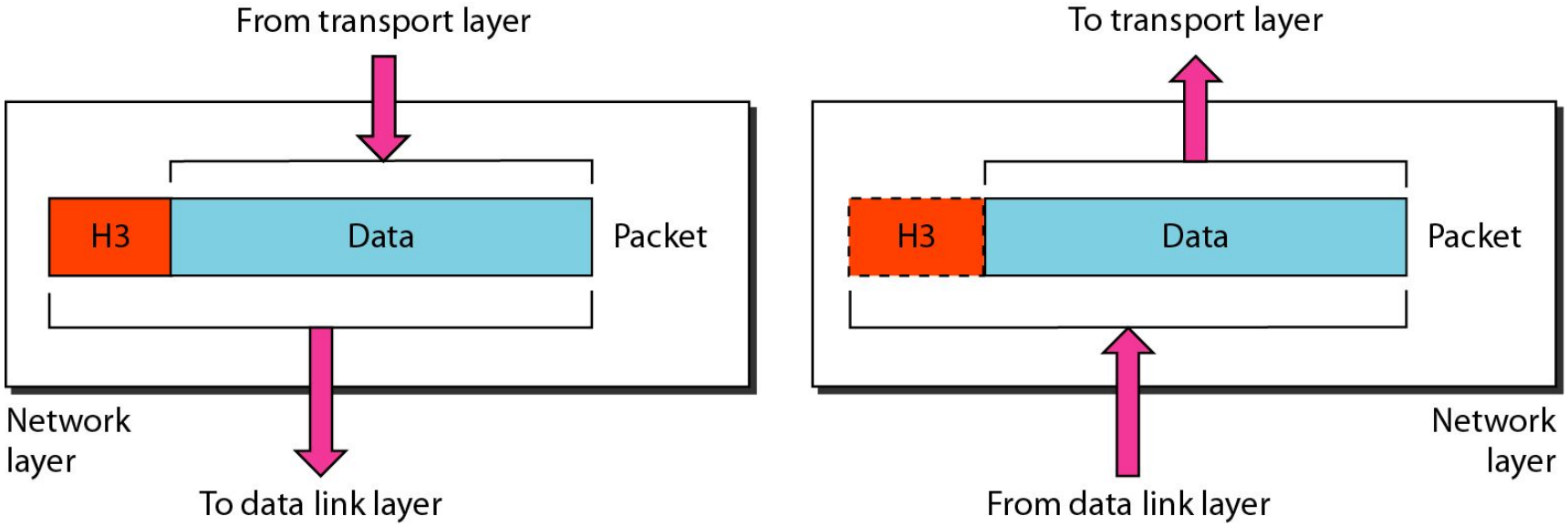
Functions– Multiplexing



NETWORK LAYER

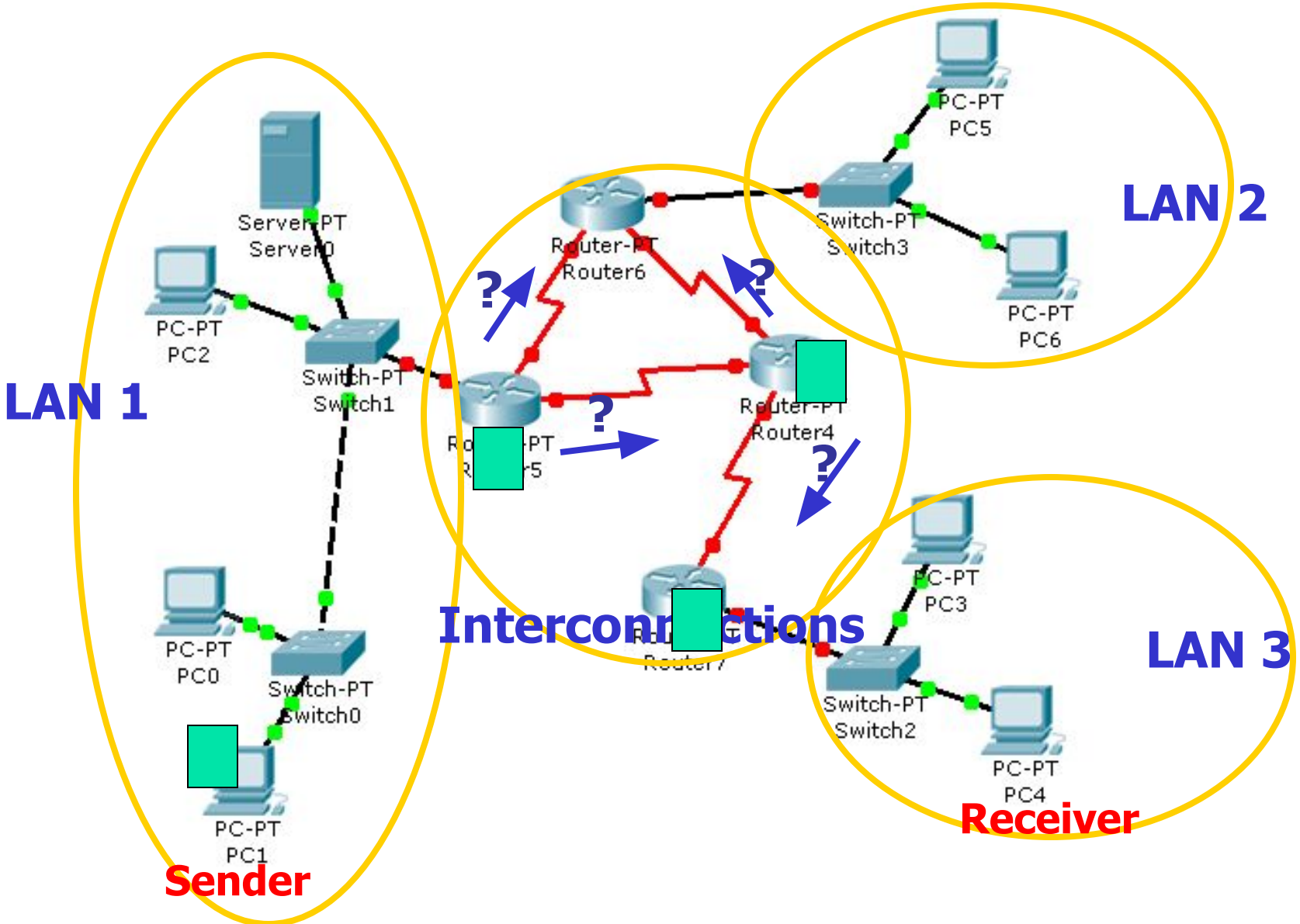
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Network layer



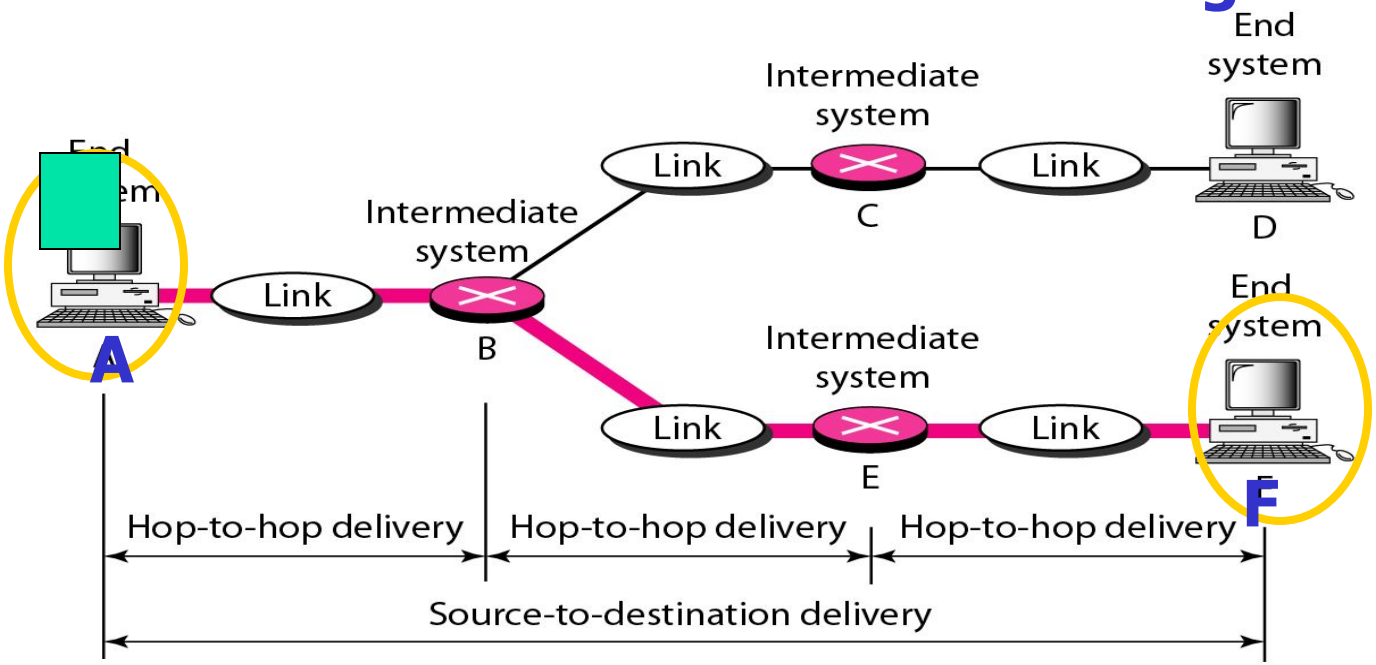
The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Example

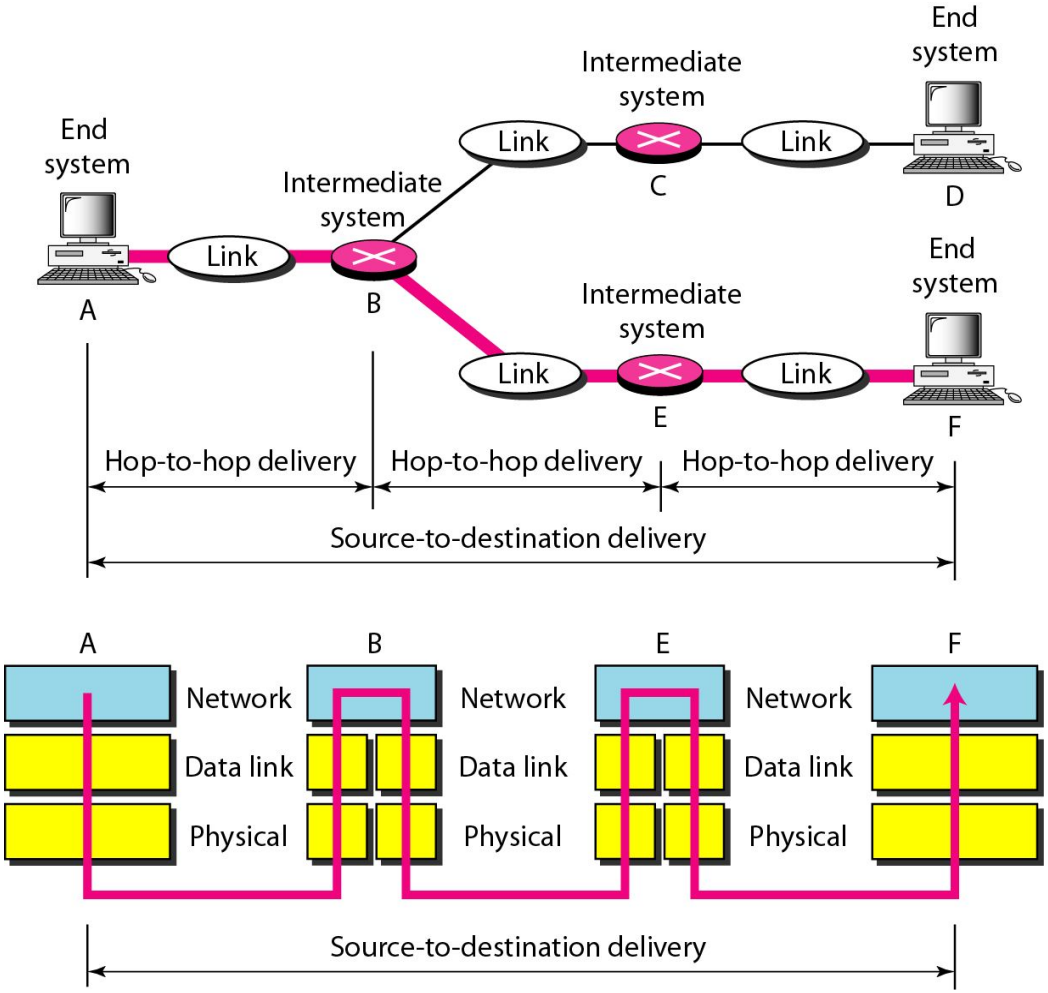

LAN 1
LAN 2
LAN 3
Interconnections
Sender
Receiver
Server-PT Server0
PC-PT PC2
Switch-PT Switch1
PC-PT PC0
Switch-PT Switch0
PC-PT PC1
Router-PT Router6
Router-PT Router4
PC-PT PC5
Switch-PT Switch3
PC-PT PC6
PC-PT PC3
Switch-PT Switch2
PC-PT PC4
?
?
?
?

Source-to-destination delivery

- Functions :
  - Adds an address to identify sender and receiver hosts. **Adds Logical Addressing**
  - Decides which path to take. **Routing**



Source-to-destination delivery



- Network Layer PDU is called **Packets**.
- Common Network layer Protocol is called **Internet Protocol (IP)**

Logical Addresses :: IP Address

- Universal address, each host uniquely defined.
- 32-bit address also known as **IP Address**.
- Independent of underlying physical networks.

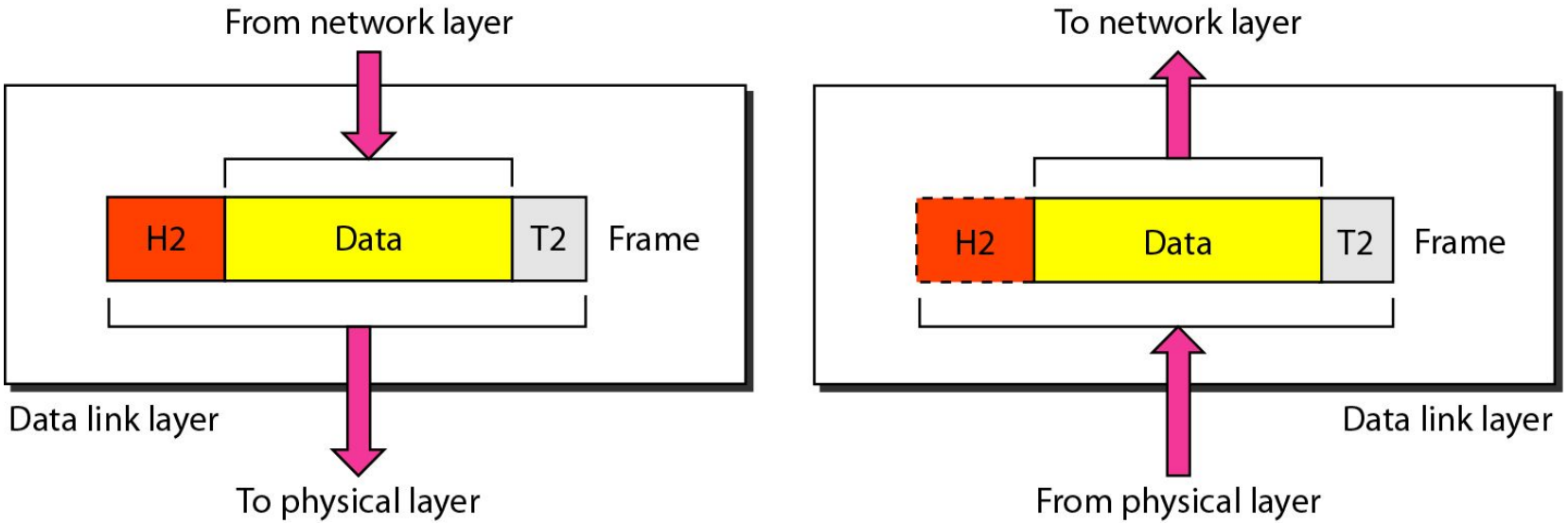
192.168.10.1

32 bits written in dotted decimal notation. Each decimal represented by 8 bits.

DATA LINK LAYER

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Data link layer



The data link layer is responsible for moving frames from one hop (node) to the next.

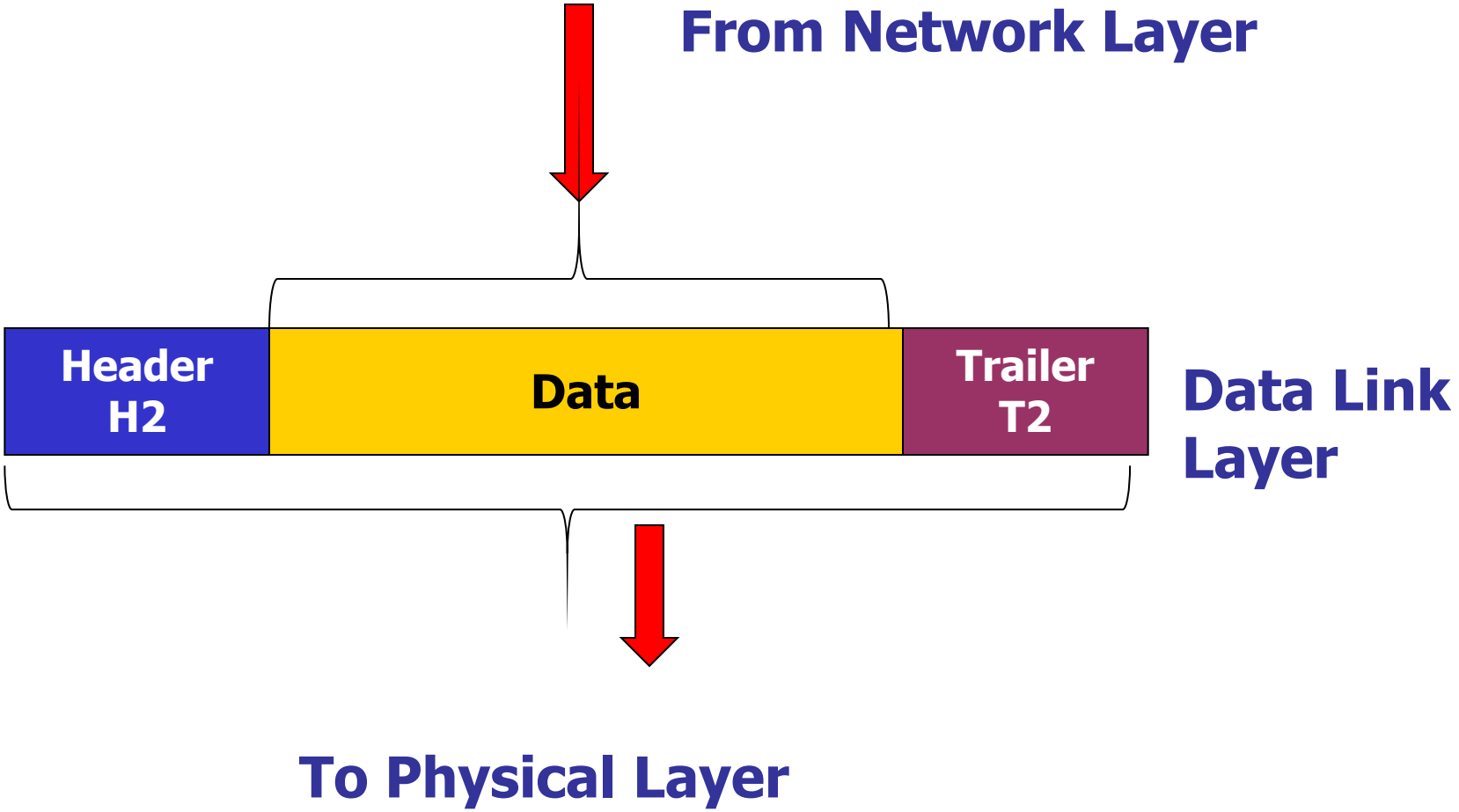
Functions-Data Link Layer

- Framing
- Physical Addressing
- Flow Control
- Error Control
- Access Control

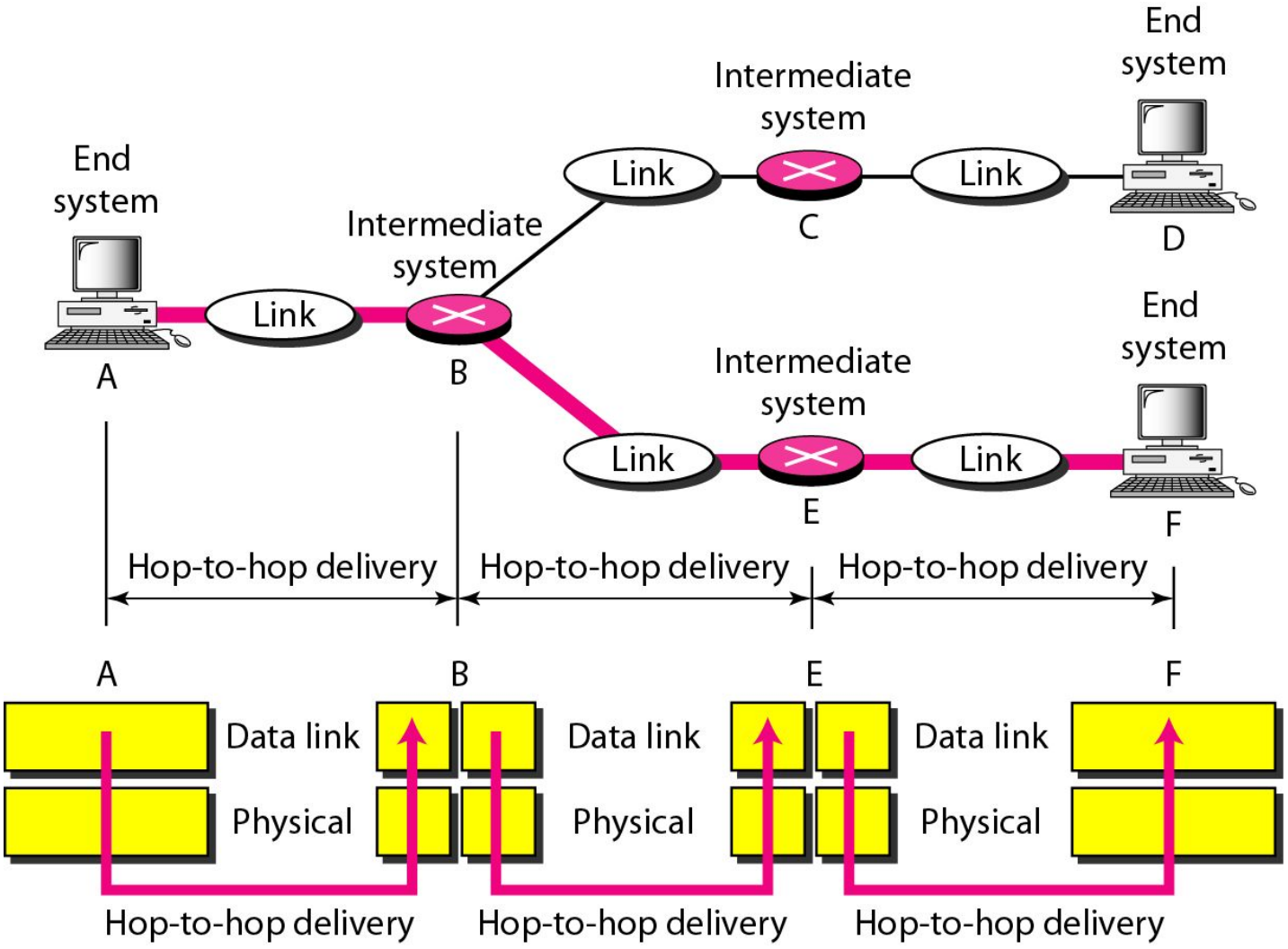
Data Link Layer PDU is called **Frames.**

Data Link Layer Protocol varies.

Functions : Framing



Hop-to-hop delivery



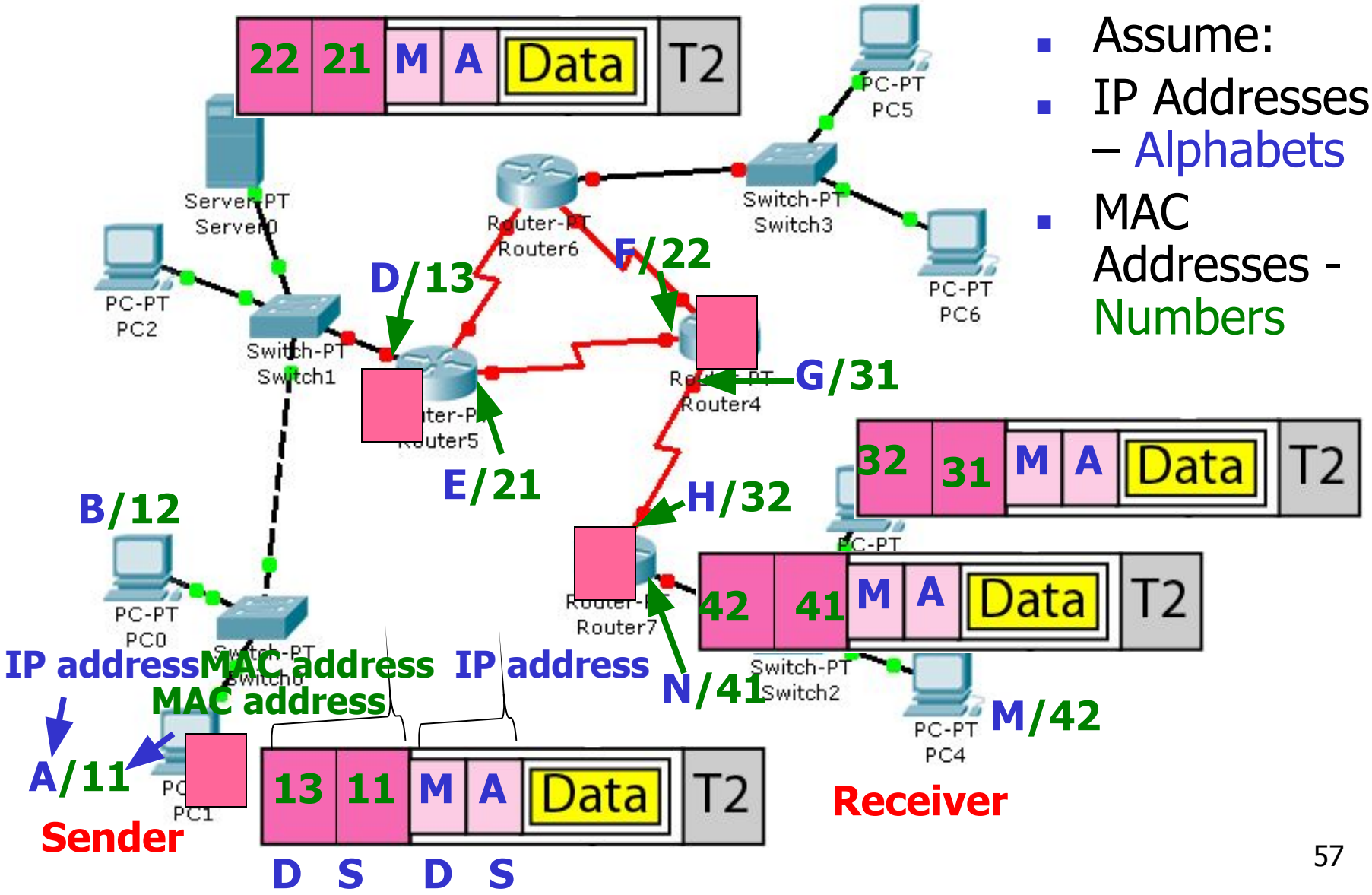
Physical Address : MAC Address

- Every interface/port has an unique identifying number.
- Given by manufacturer.
- 48 bits long, represented by 12 hexadecimal digits.

07:01:02:01:2C:4B

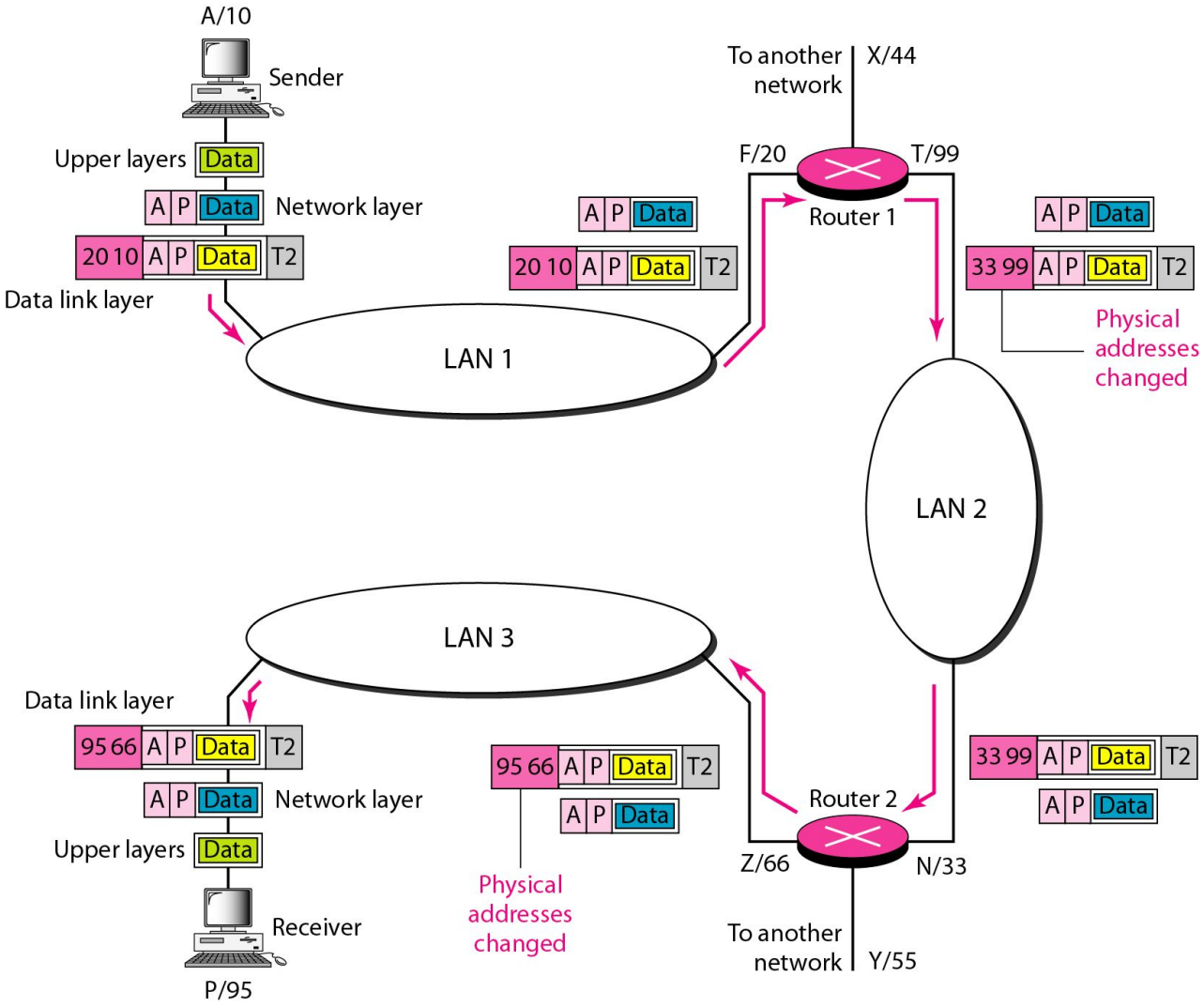
Also known as MAC (Media Access Control) Address.

Addressing

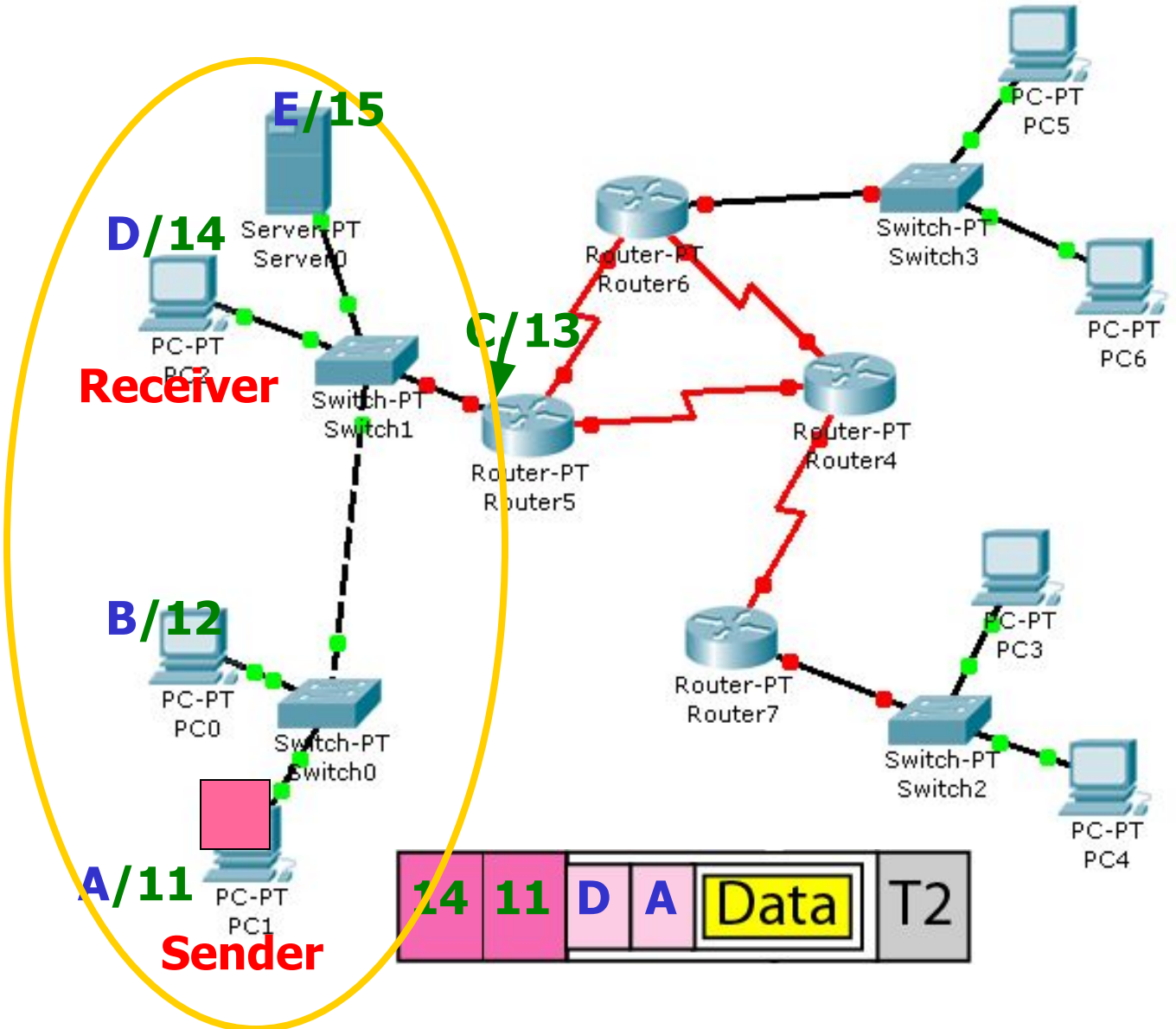


- Assume:
- IP Addresses – Alphabets
- MAC Addresses - Numbers

Addressing


A/10
Sender
Upper layers
Data
A P Data
Network layer
20 10 A P Data T2
Data link layer
LAN 1
To another network
X/44
F/20
T/99
Router 1
A P Data
20 10 A P Data T2
33 99 A P Data T2
Physical addresses changed
LAN 2
LAN 3
Data link layer
95 66 A P Data T2
A P Data
Network layer
Upper layers
Data
Receiver
P/95
95 66 A P Data T2
A P Data
Physical addresses changed
Router 2
Z/66
N/33
33 99 A P Data T2
A P Data
To another network
Y/55

Addressing-Within the same network

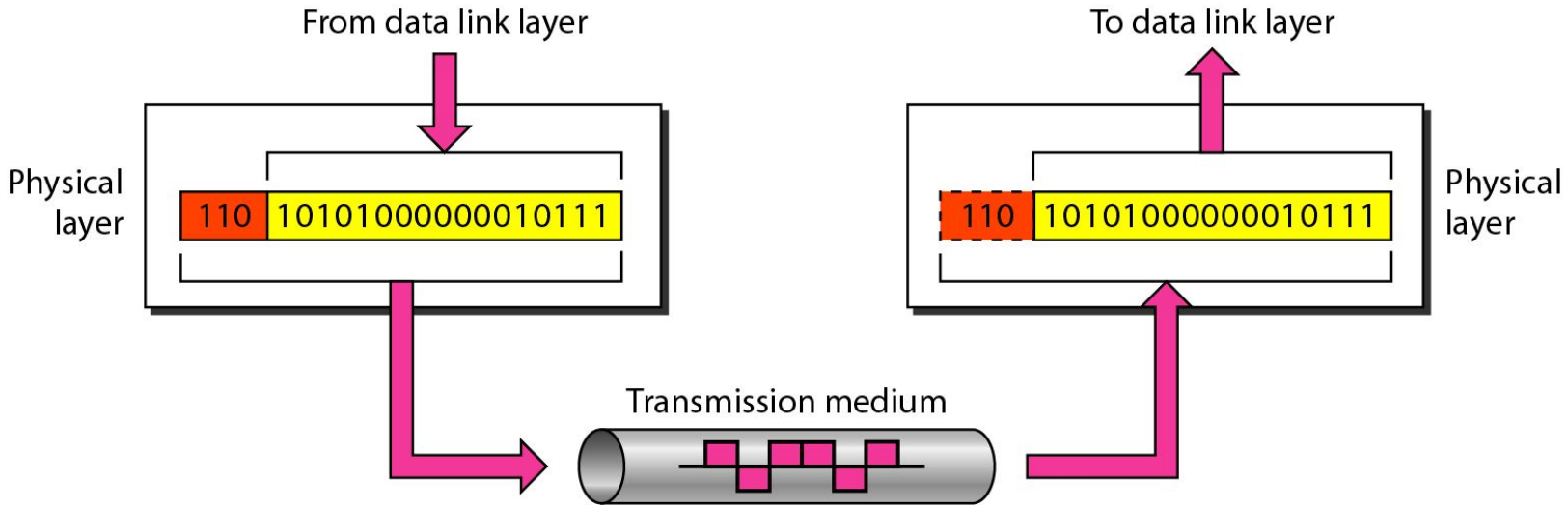


- Assume:
- IP Addresses – Alphabets
- MAC Addresses - Numbers

PHYSICAL LAYER

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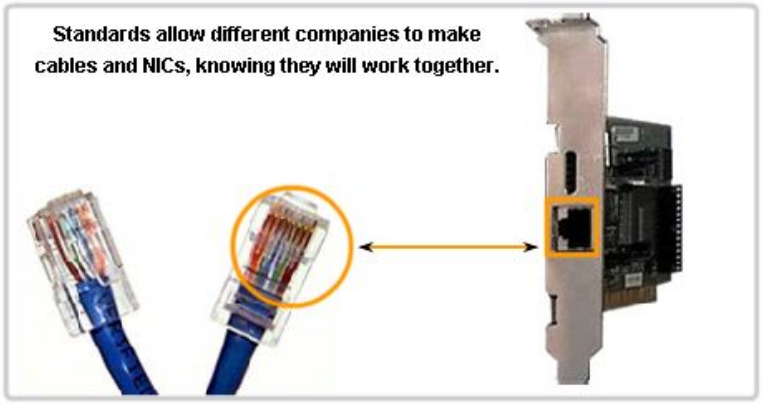
Physical layer

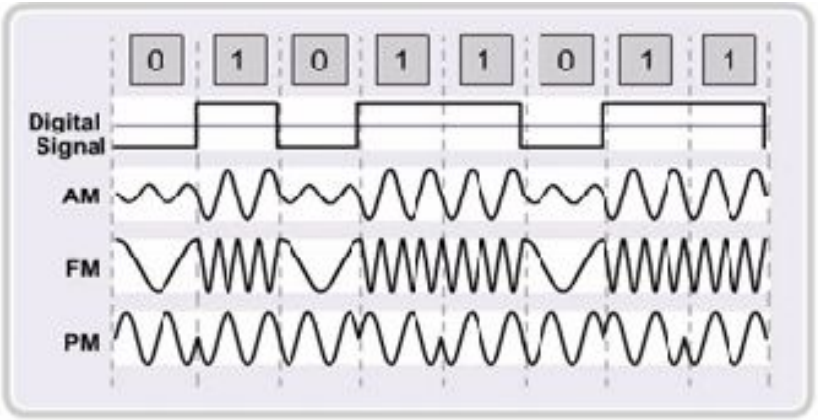


The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Functions-Physical Layer

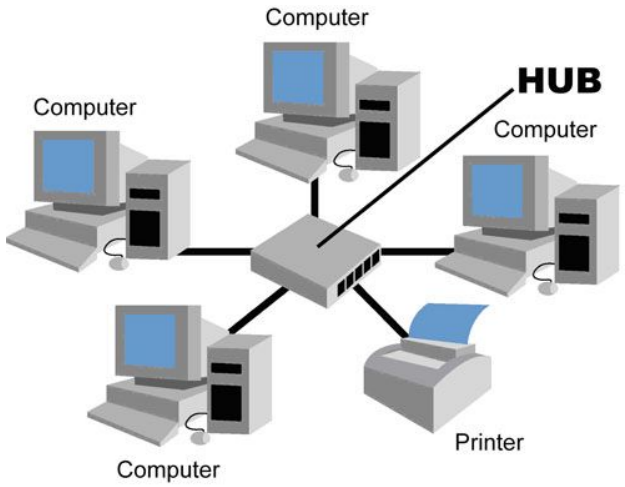
- Physical Characteristics of interfaces and medium.
- Representation of bits
- Data Rate
- Synchronization of bits

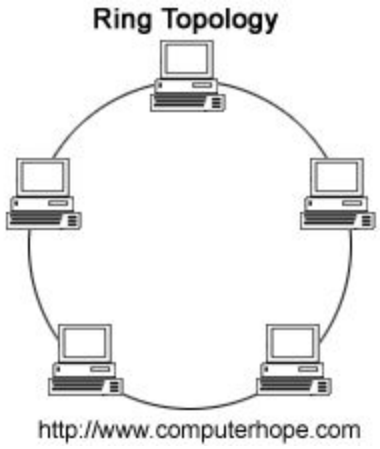




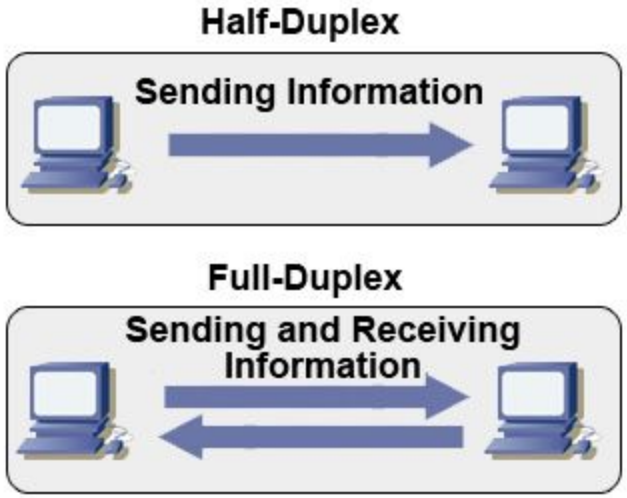
Functions-Physical Layer

- Physical Topology
  - Bus
  - Ring etc

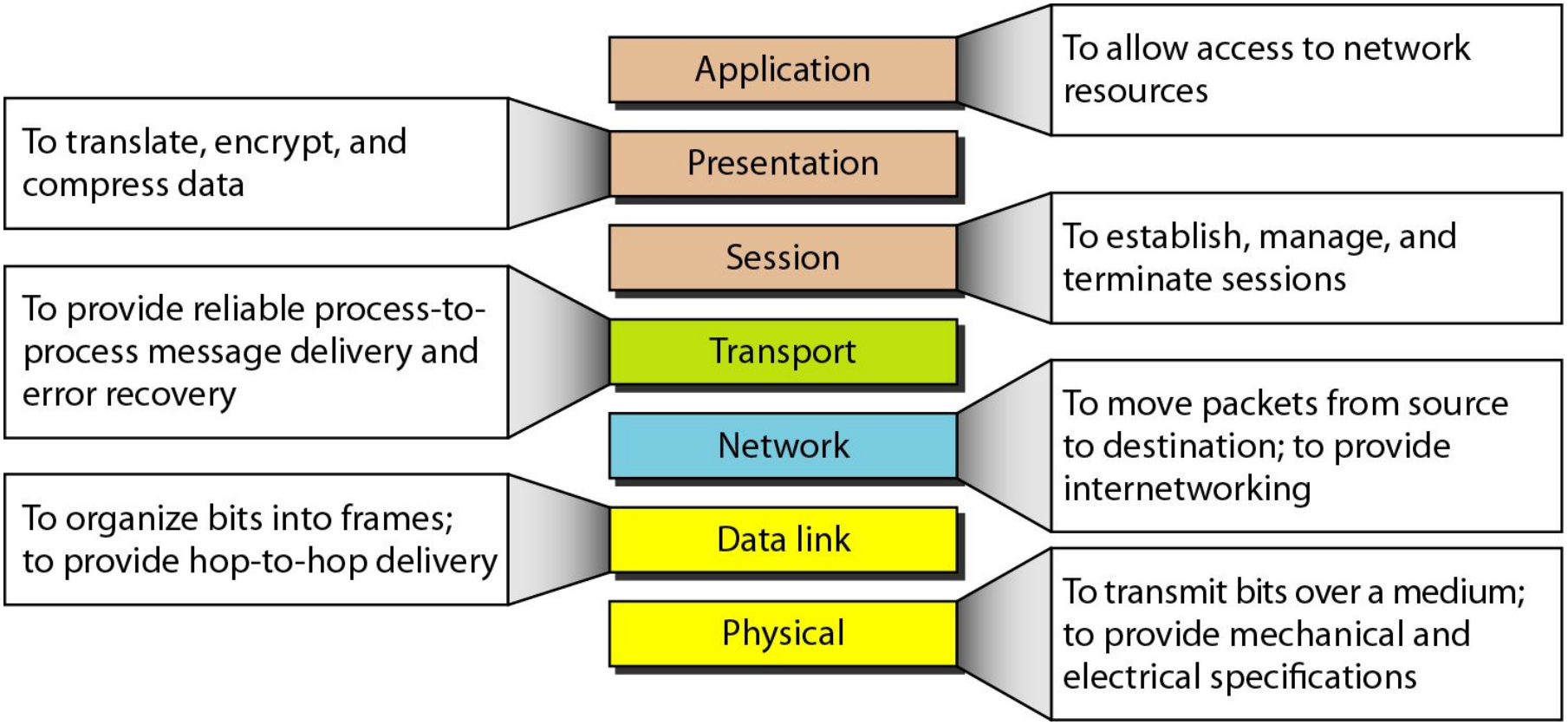




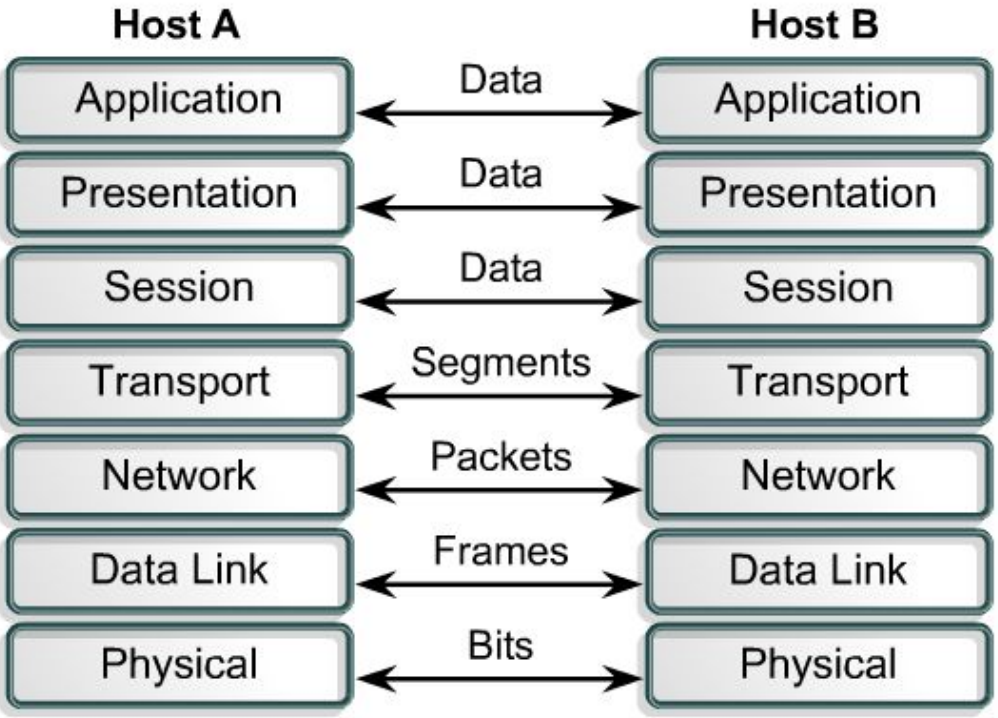
- Transmission Modes
  - Simplex
  - Half Duplex
  - Full Duplex



Summary of OSI Layers



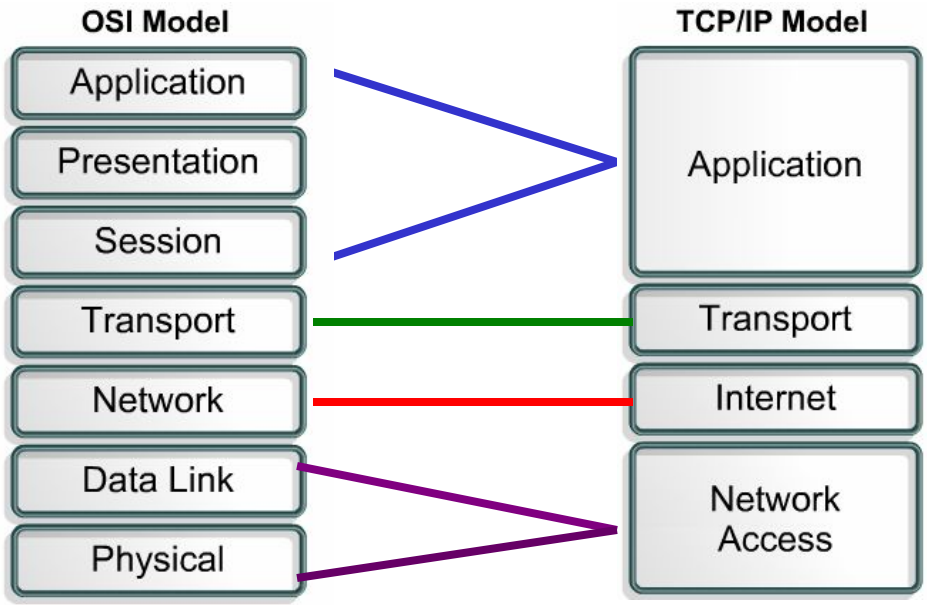
Summary



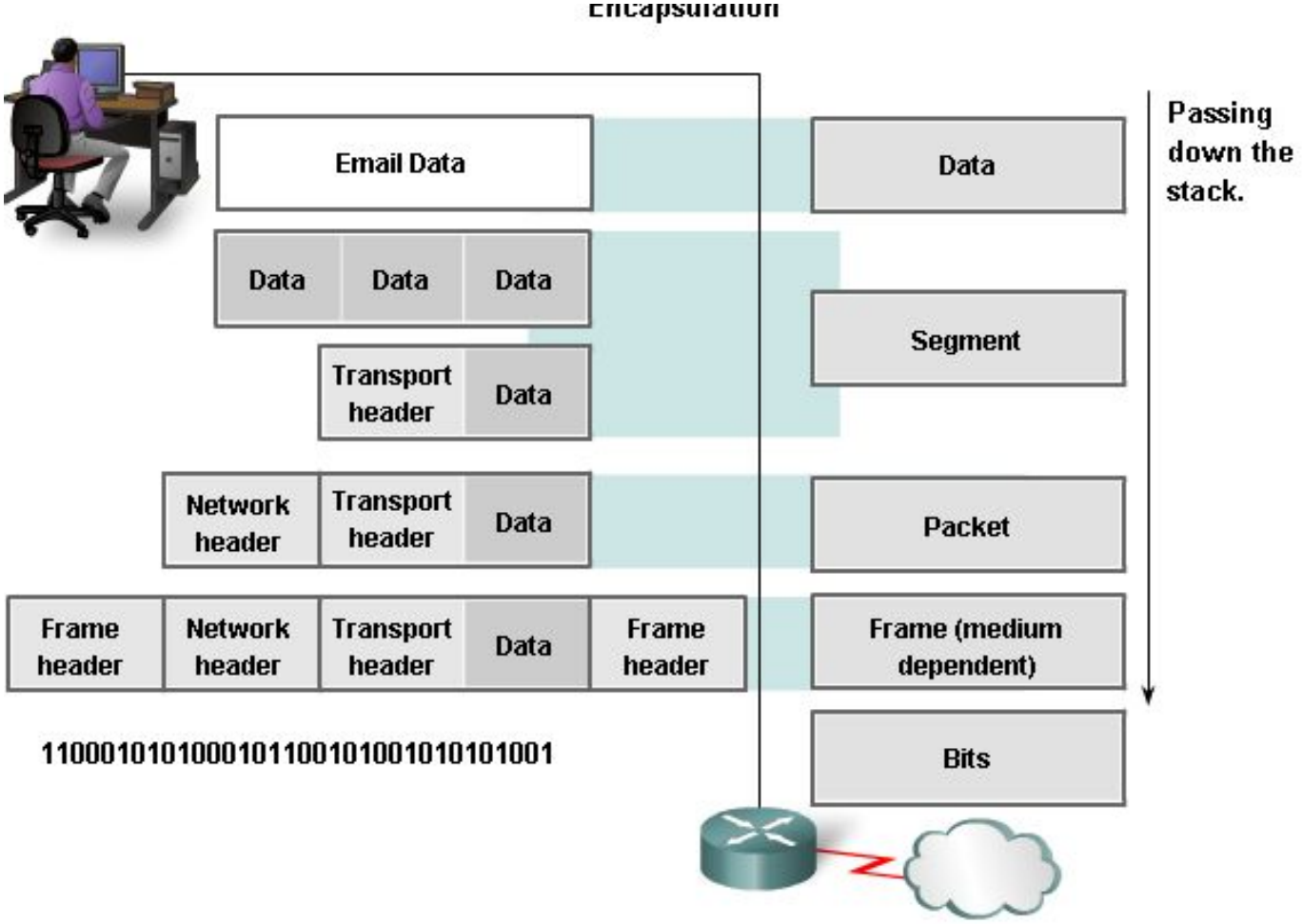
- PDUs
- Encapsulation
- Headers and trailers

TCP/IP

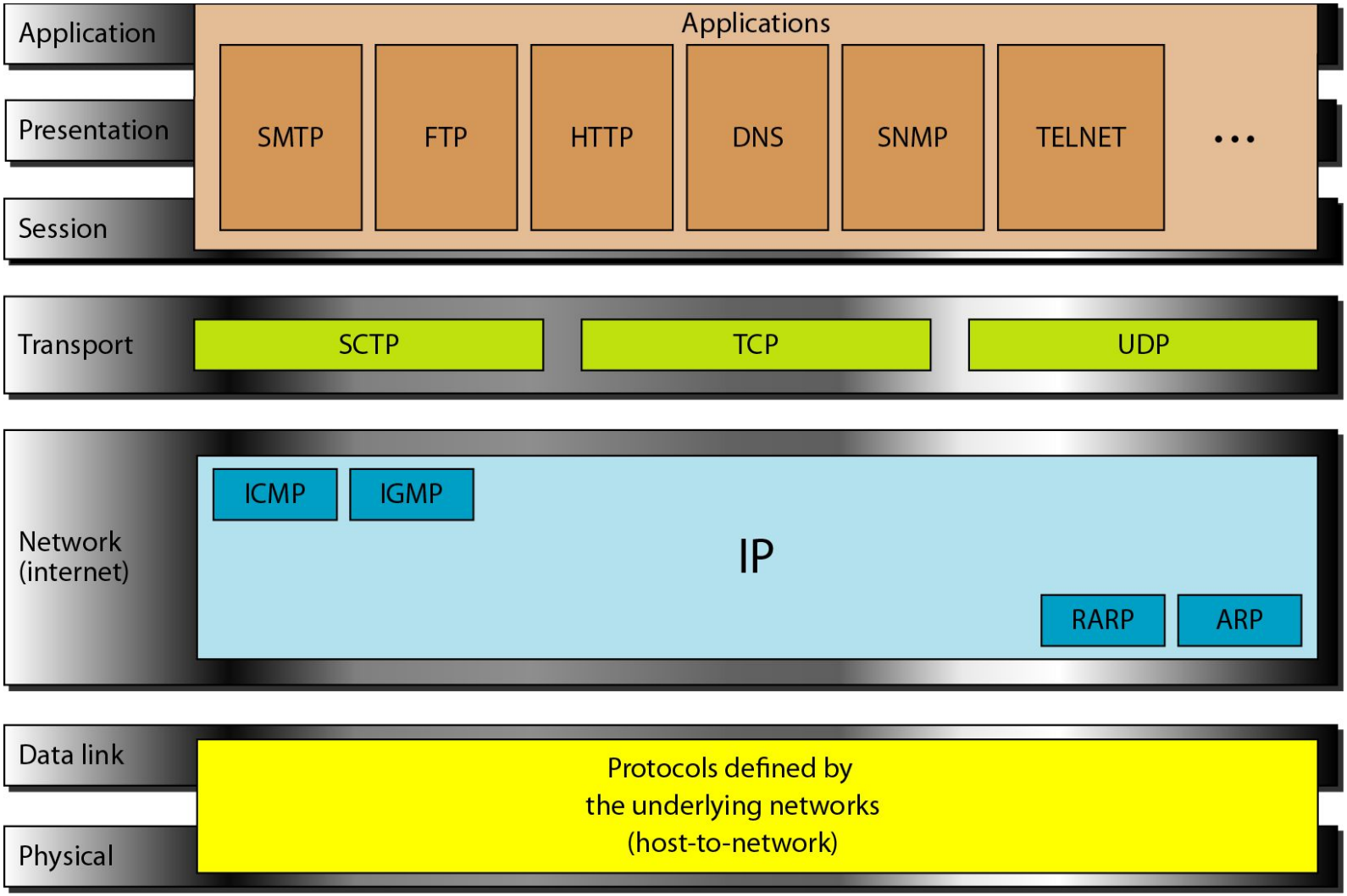
- Developed by the US Defense Advanced Research Project Agency (DARPA) for its packet switched network (ARPANET)
- Used by the global Internet.
- De Facto Standard



TCP/IP Encapsulation

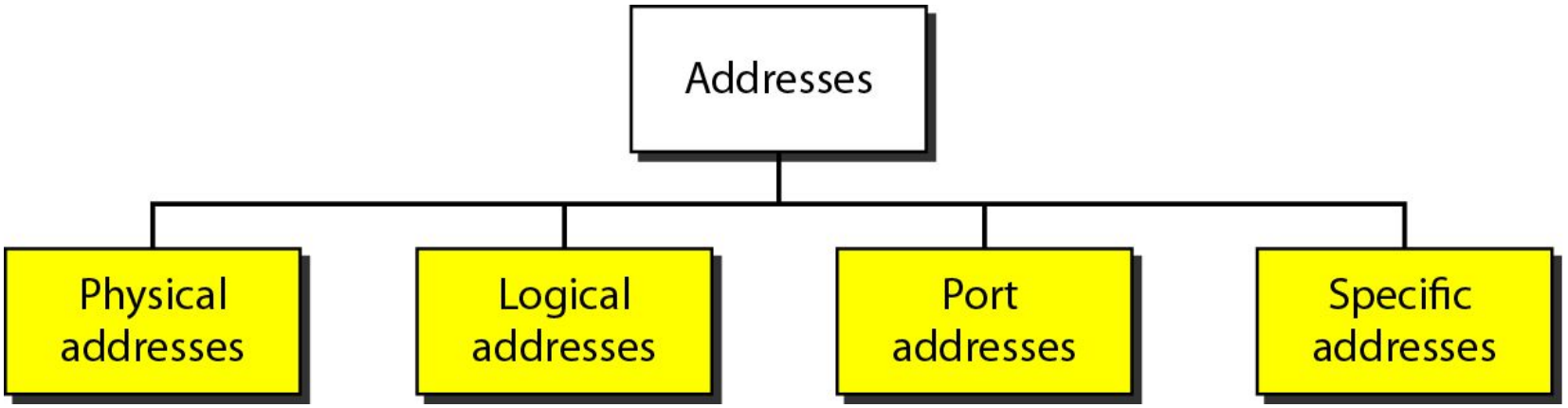


TCP/IP and OSI model

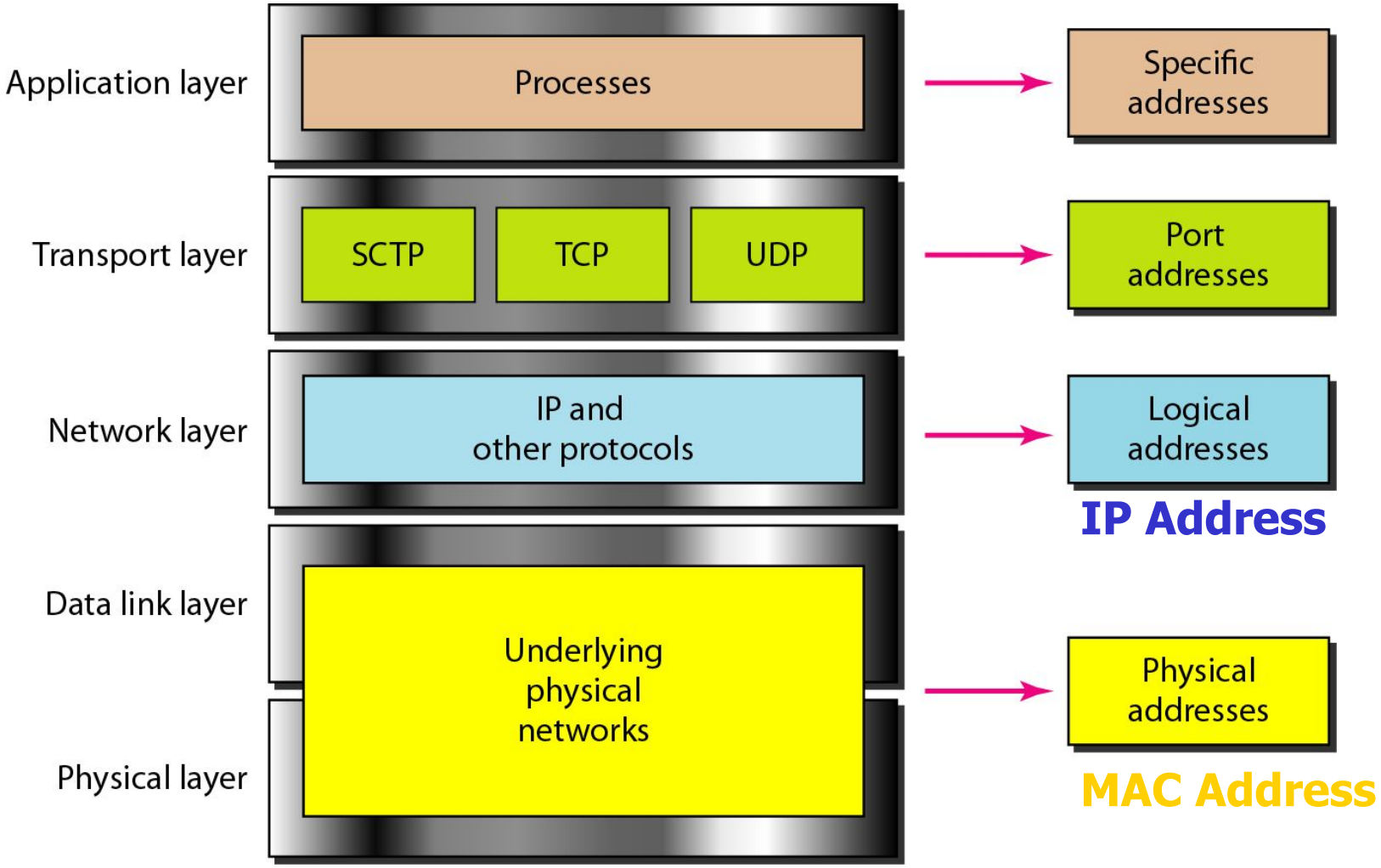


Addressing Summary

- Four levels of addresses are used in an internet employing the TCP/IP protocols



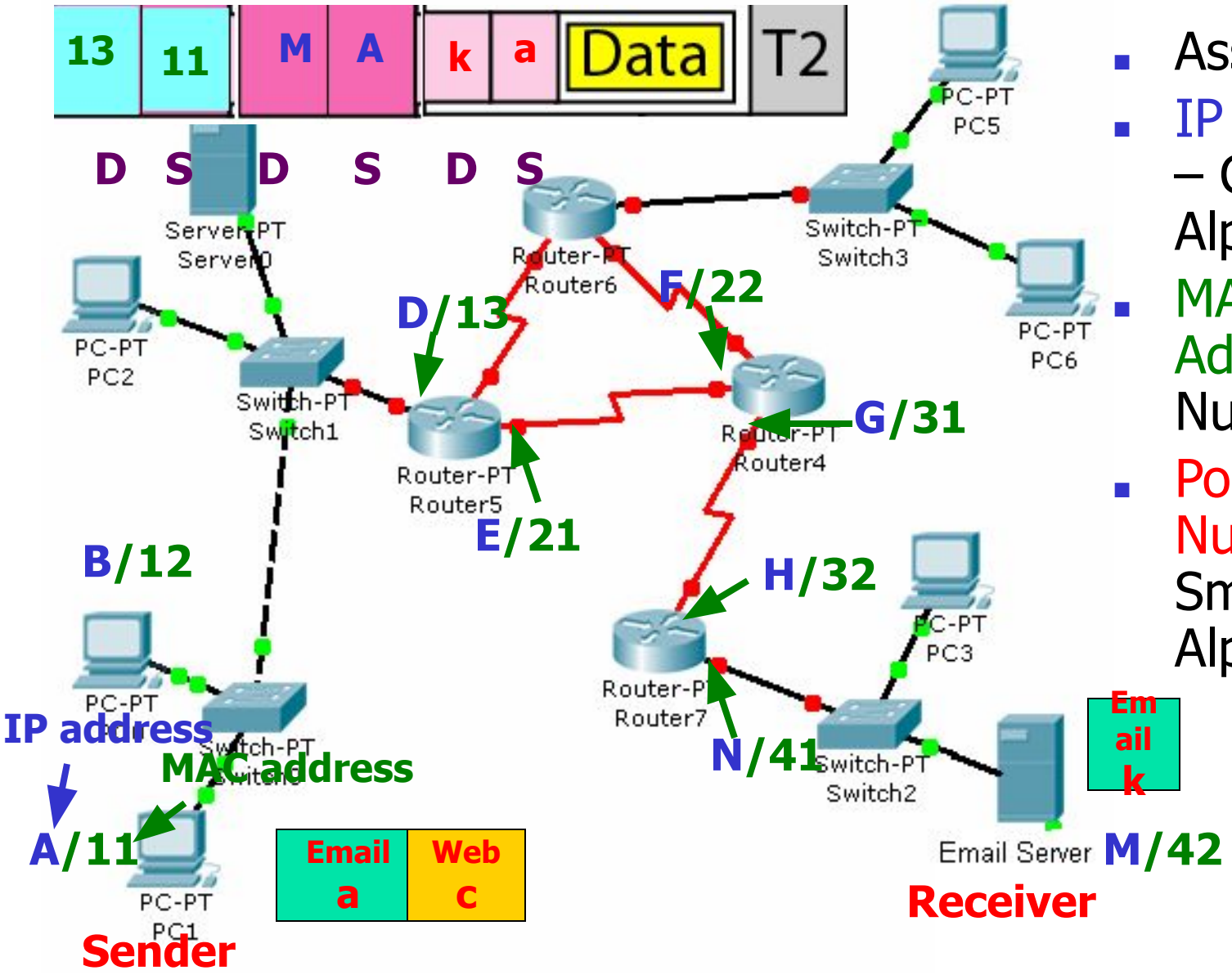
Relationship of layers and addresses in TCP/IP



Specific Addresses

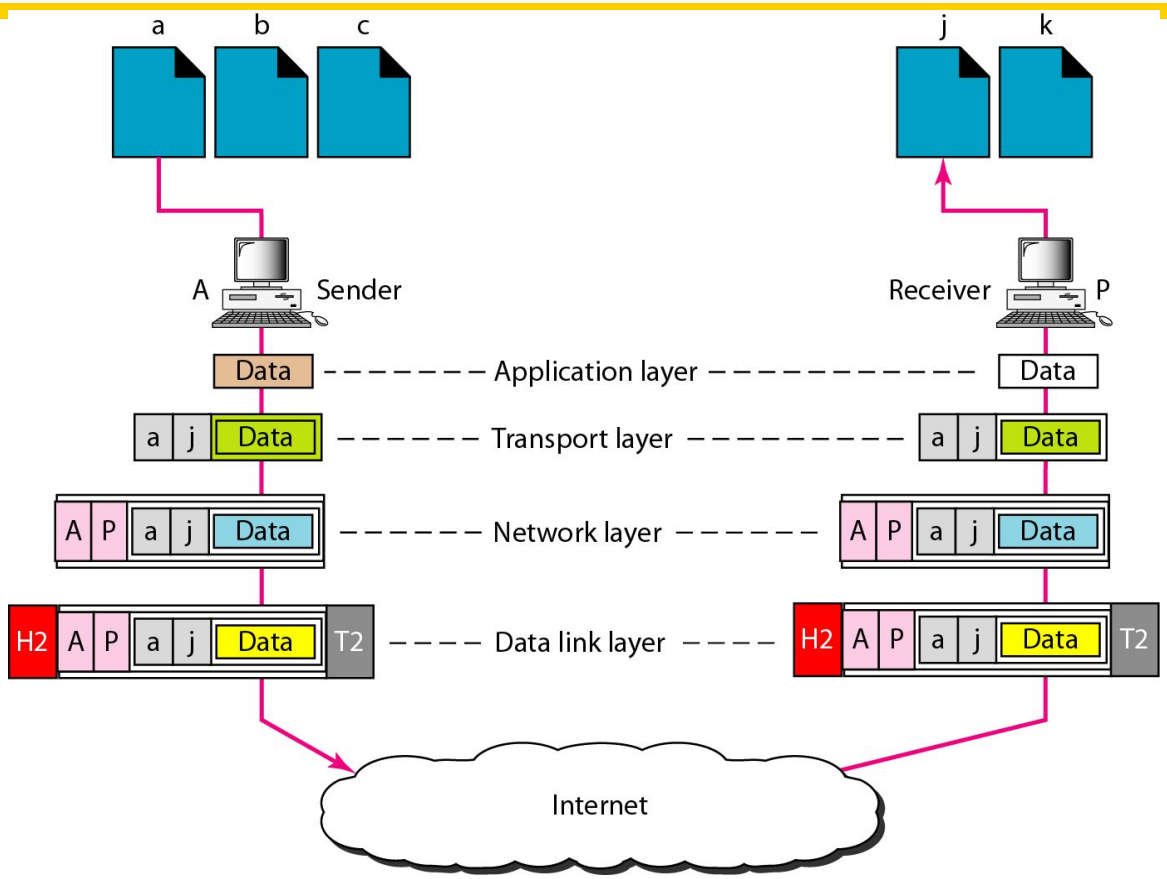
- Applications having user friendly addresses.
- Email addresses or URLs.
  - john@gmail.comjohn@gmail.com or www.bracu.ac.bd
- These are converted into corresponding port and logical addresses by the sending computer.

Addressing Review



- Assume:
- IP Addresses – Capital Alphabets
- MAC Addresses – Numbers
- Port Numbers- Small Alphabets

Addressing Review



Although physical addresses change from hop to hop, logical and port addresses remain the same from the source to destination.

END

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